

SAPIENZA UNIVERSITÀ DI ROMA

18 MARCH 2026

DETECTION & CHARACTERIZATION OF GRAVITATIONAL-WAVE BURSTS WITH MULTI-DETECTOR NETWORKS

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Talk Overview

INTRODUCTION

Gravitational-wave bursts
&
Instrumental
glitches

BayesWave:
algorithm
overview

FOCUSSED TOPICS

Multi-detector
impact on
BayesWave's
**detection
performance**

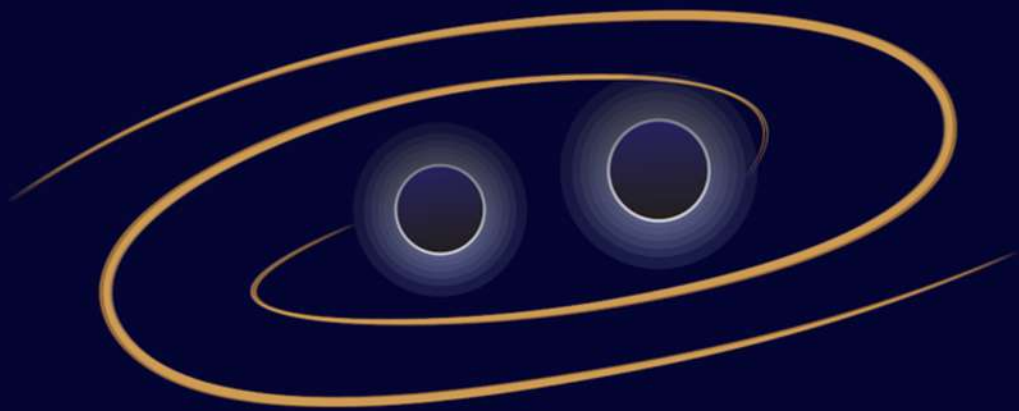
Dedicated-
frequency
analysis of
**Core-collapse
Supernovae**

Characterizing
gravitational-
wave burst
**tensor
polarizations**

Types of Gravitational Waves

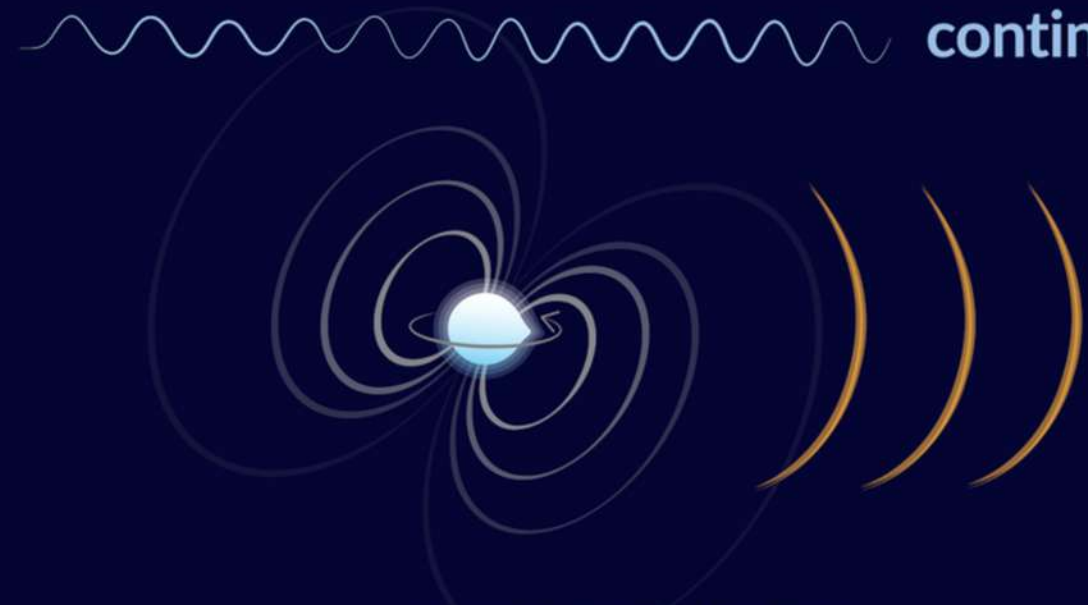
$\mathcal{O}(10^2)$
events
detected

compact binary inspiral



MODELLED

continuous



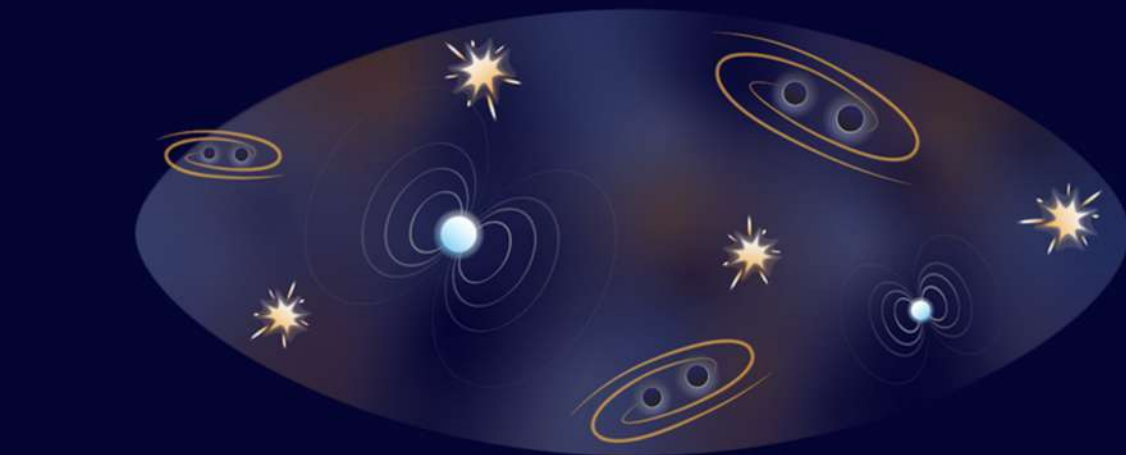
SHORT DURATION

LONG DURATION

UNMODELLED



burst

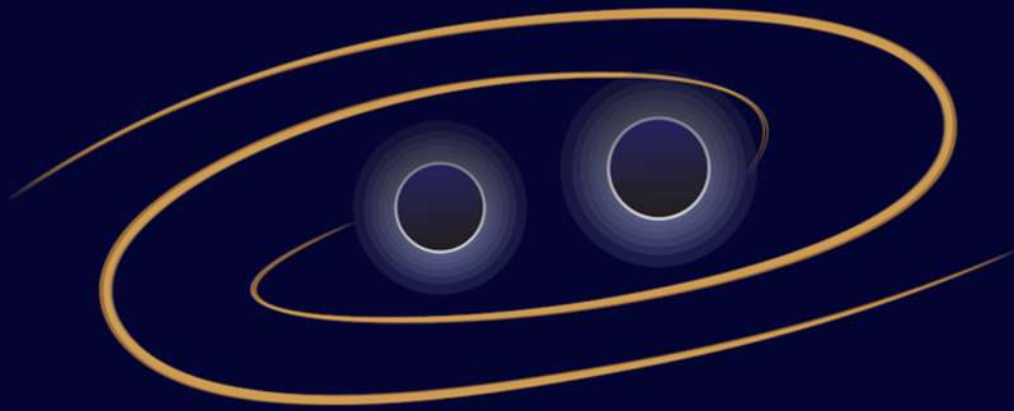


stochastic

@astronerdika

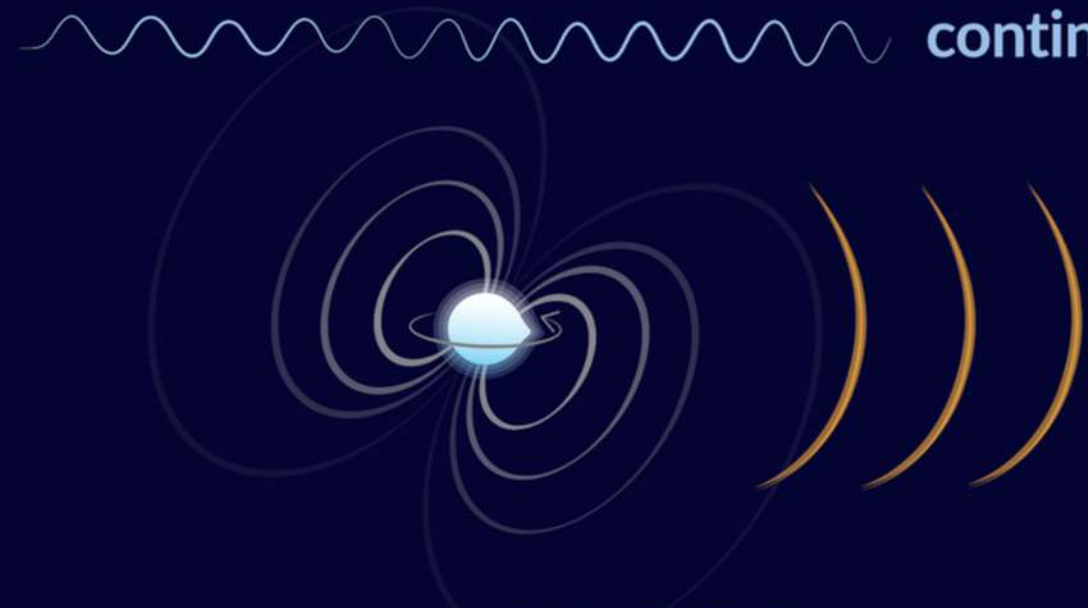
Types of Gravitational Waves

compact binary inspiral



MODELLED

continuous



SHORT DURATION

LONG DURATION

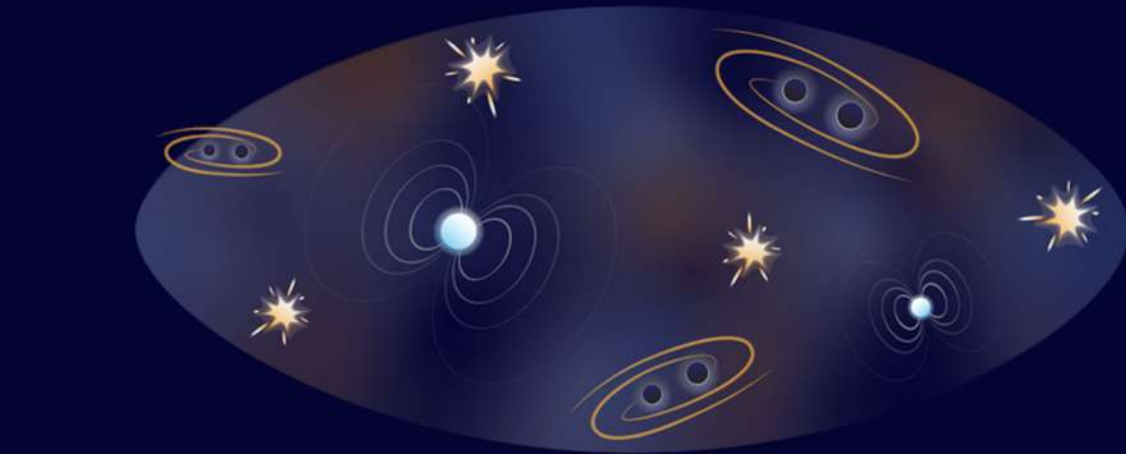
Focus of
this talk

burst



UNMODELLED

stochastic



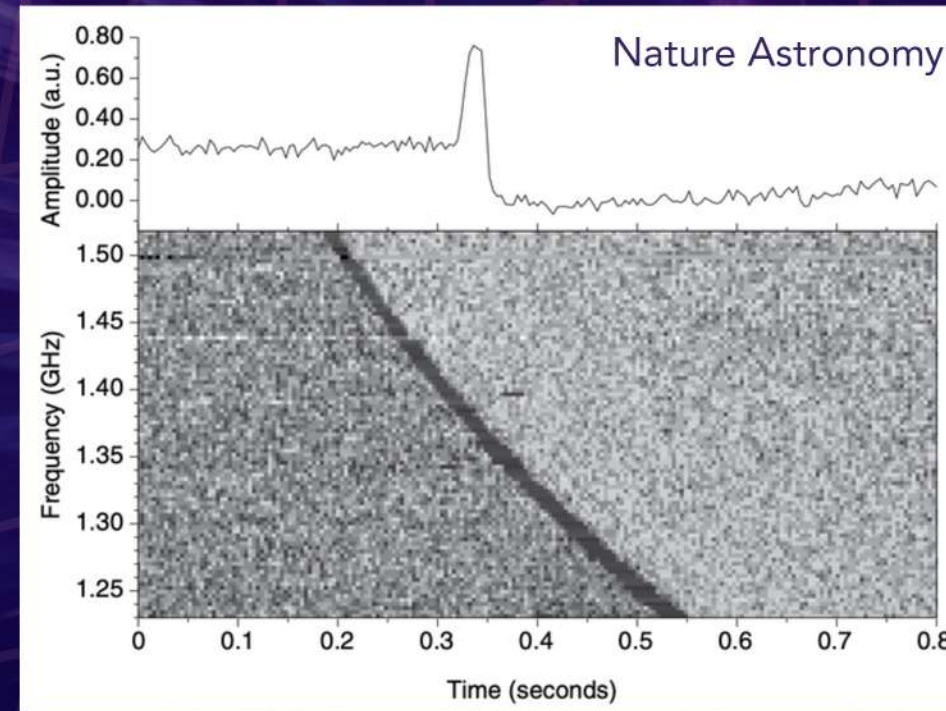
@astronerdika

Why search for **Unmodelled bursts?**

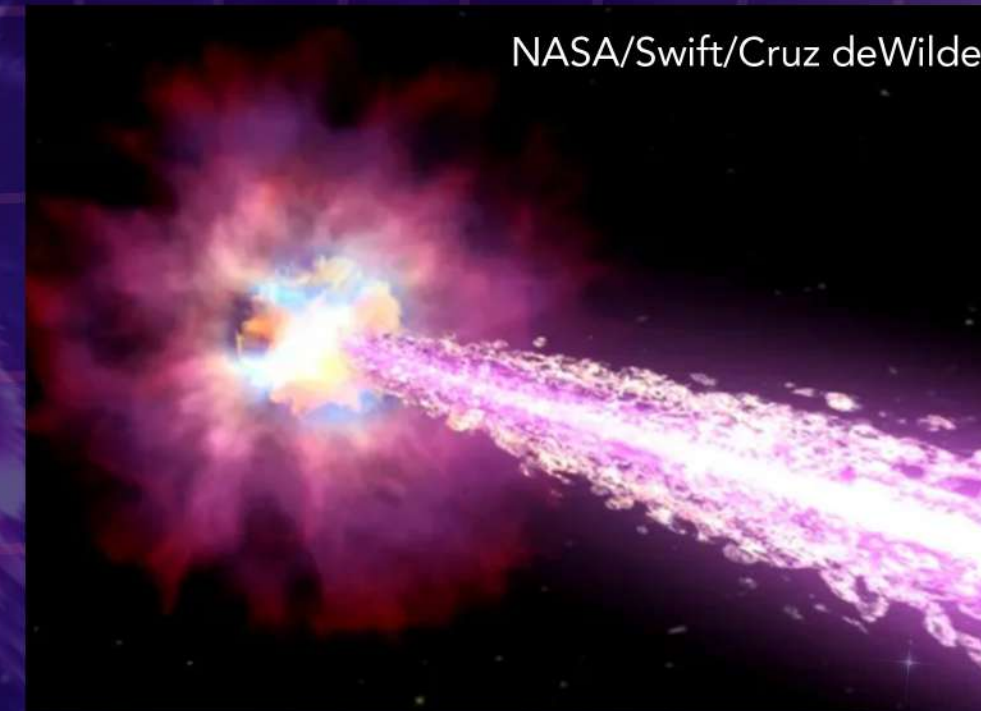
Multi-messenger astronomy for known sources



Core-collapse
supernovae (CCSNe)



Fast radio bursts



Gamma ray bursts



The unknown unknown:
Potential discovery of new, obscure sources



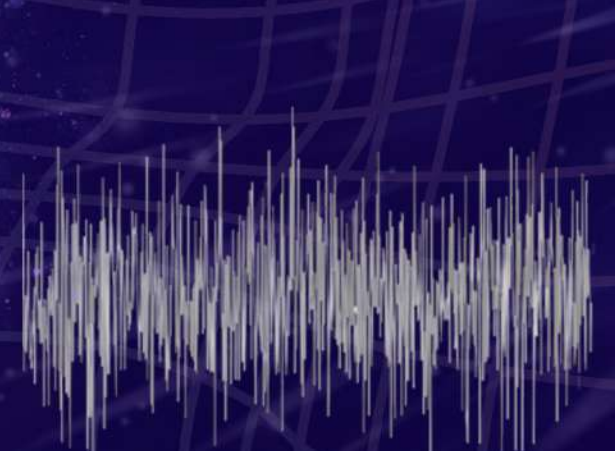
What is in the **GW** detector data



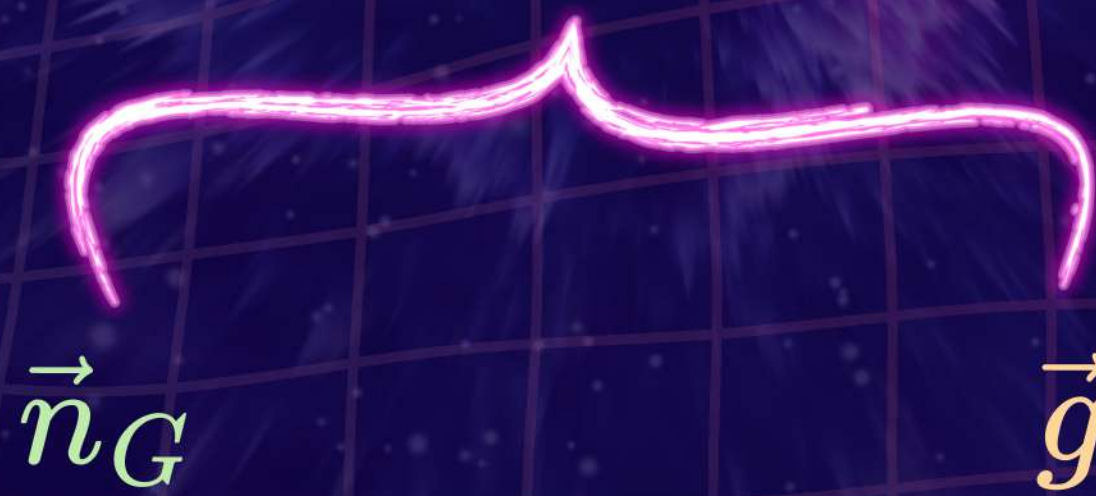
The diagram illustrates the decomposition of GW detector data into noise and signal components. It features three semi-circular arcs representing different parts of the data. The first arc, labeled $\vec{s}$ and **Data**, is solid blue. The second arc, labeled $\vec{n}$ and **Noise**, is divided into a light blue segment and a dark blue segment. The third arc, labeled $\vec{h}(\vec{\theta})$ and **Signal**, is also divided into a light blue segment and a dark blue segment. An equals sign is placed between the first and second arcs, and a plus sign is placed between the second and third arcs, indicating the equation $\vec{s} = \vec{n} + \vec{h}(\vec{\theta})$.

$$\vec{s} = \vec{n} + \vec{h}(\vec{\theta})$$

Data **Noise** **Signal**

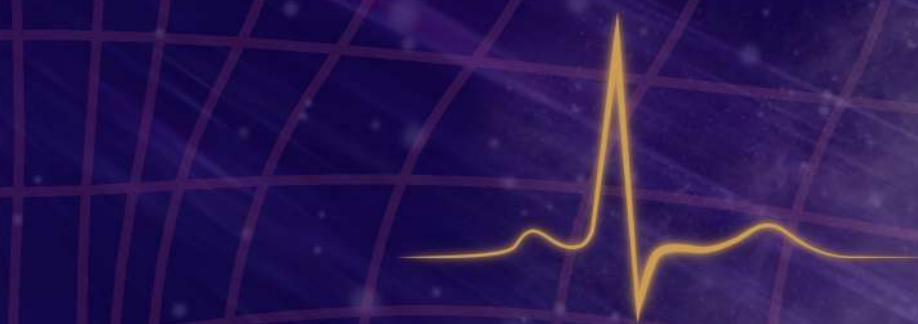


Stationary,
Gaussian noise



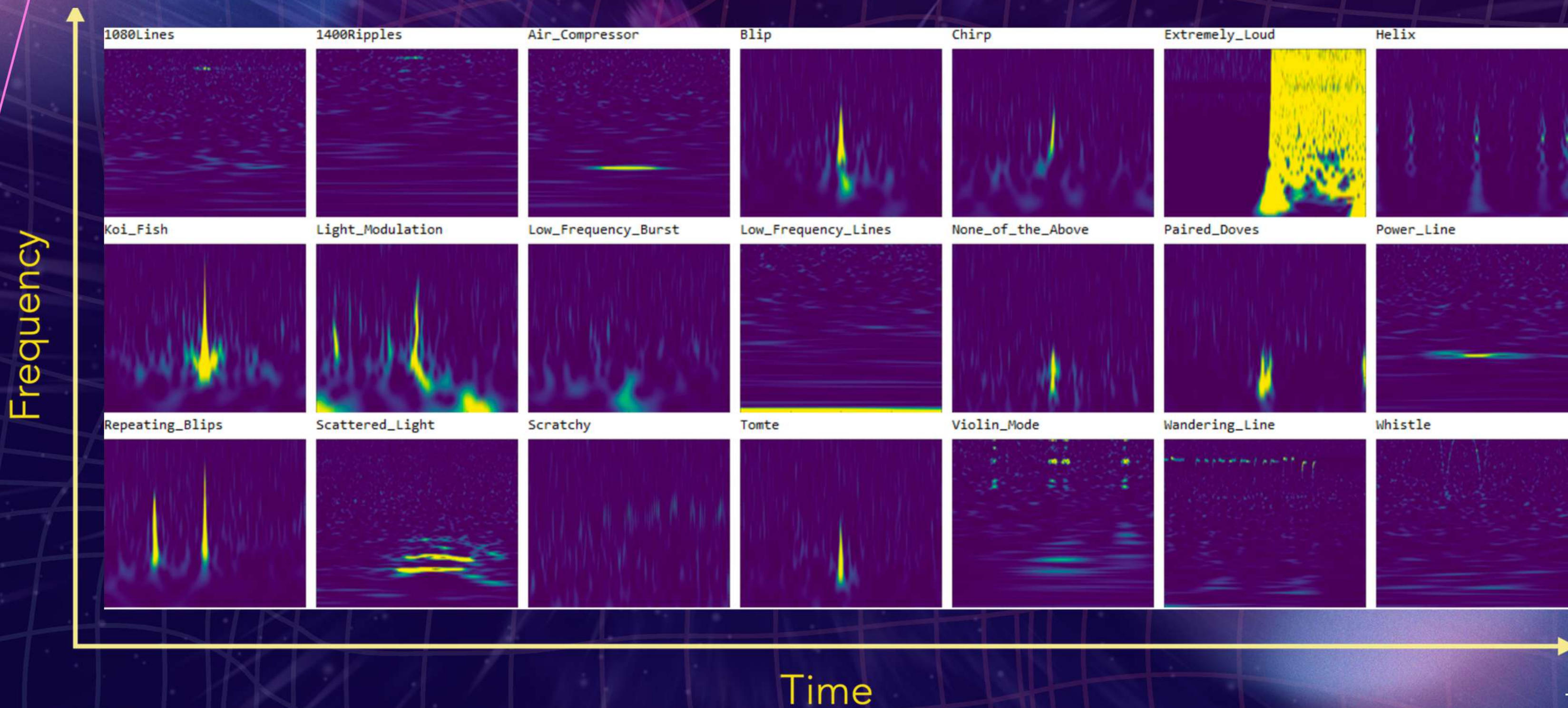
$\vec{n}_G$

$\vec{g}$



Non-stationary, non-Gaussian,
Instrumental glitch(es)

The problem: Instrumental glitches



A solution:

The *BayesWave* algorithm

Joint detection/characterization of
burst signals & glitches
with minimal assumptions

Bayes → Bayesian detection statistic

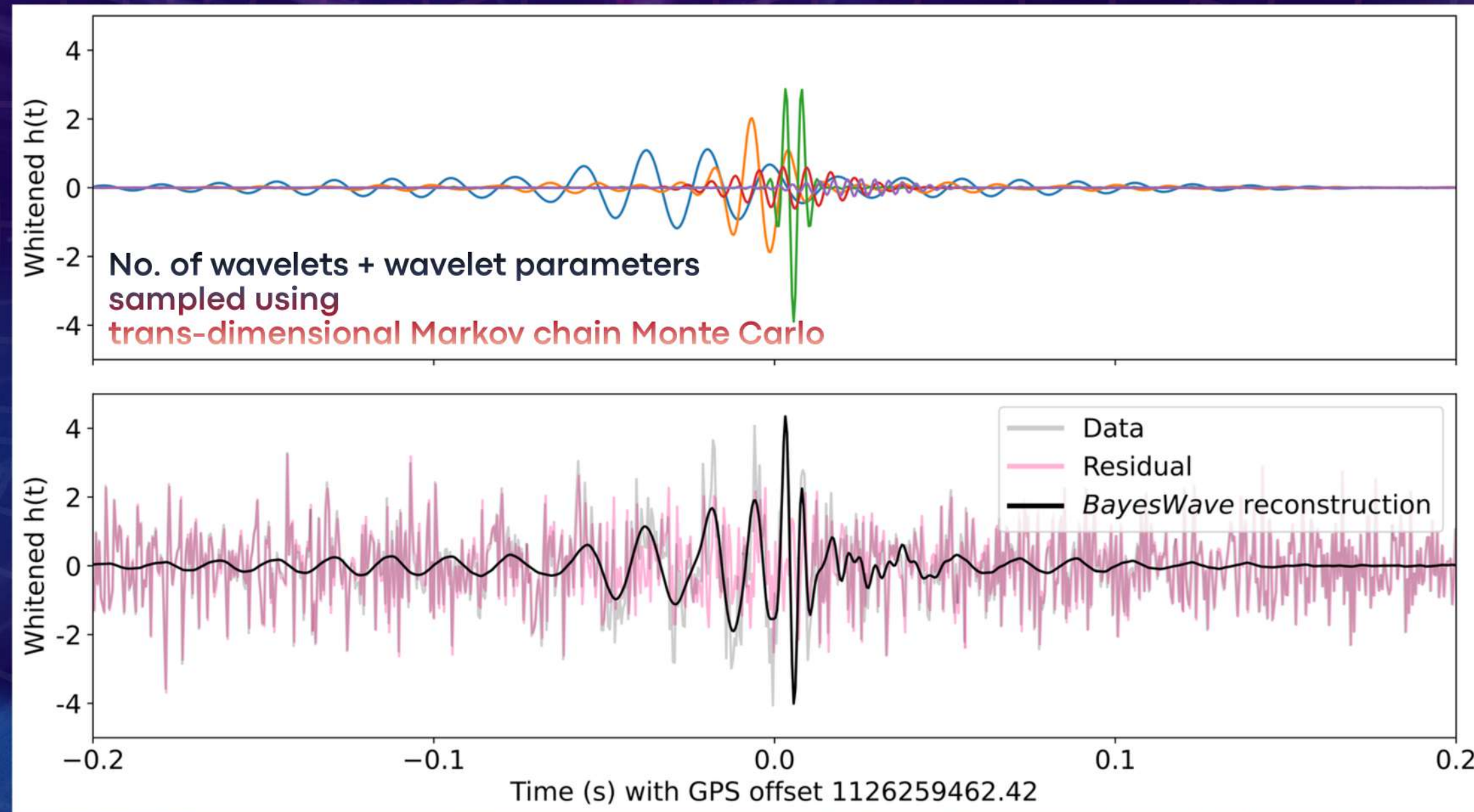
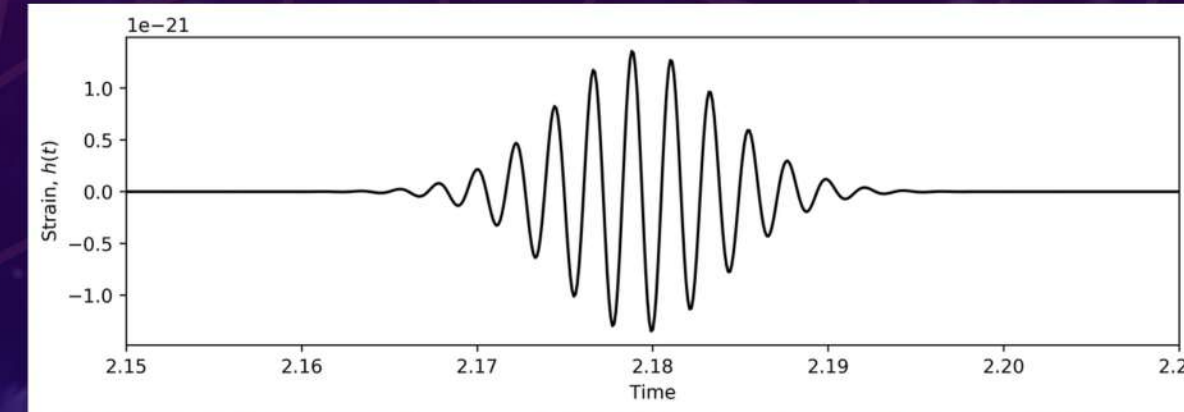
Wave → Wavelets as reconstruction frames

BayesWave publications:

Cornish et al., Phys. Rev. D 103, 044006 (2021)

Cornish + Littenberg, Class. Quant. Grav 32, 130512 (2015)

Reconstruction with **Sine-Gaussian** wavelets



EXAMPLE

Signal
reconstruction
for GW150914

Model selection:

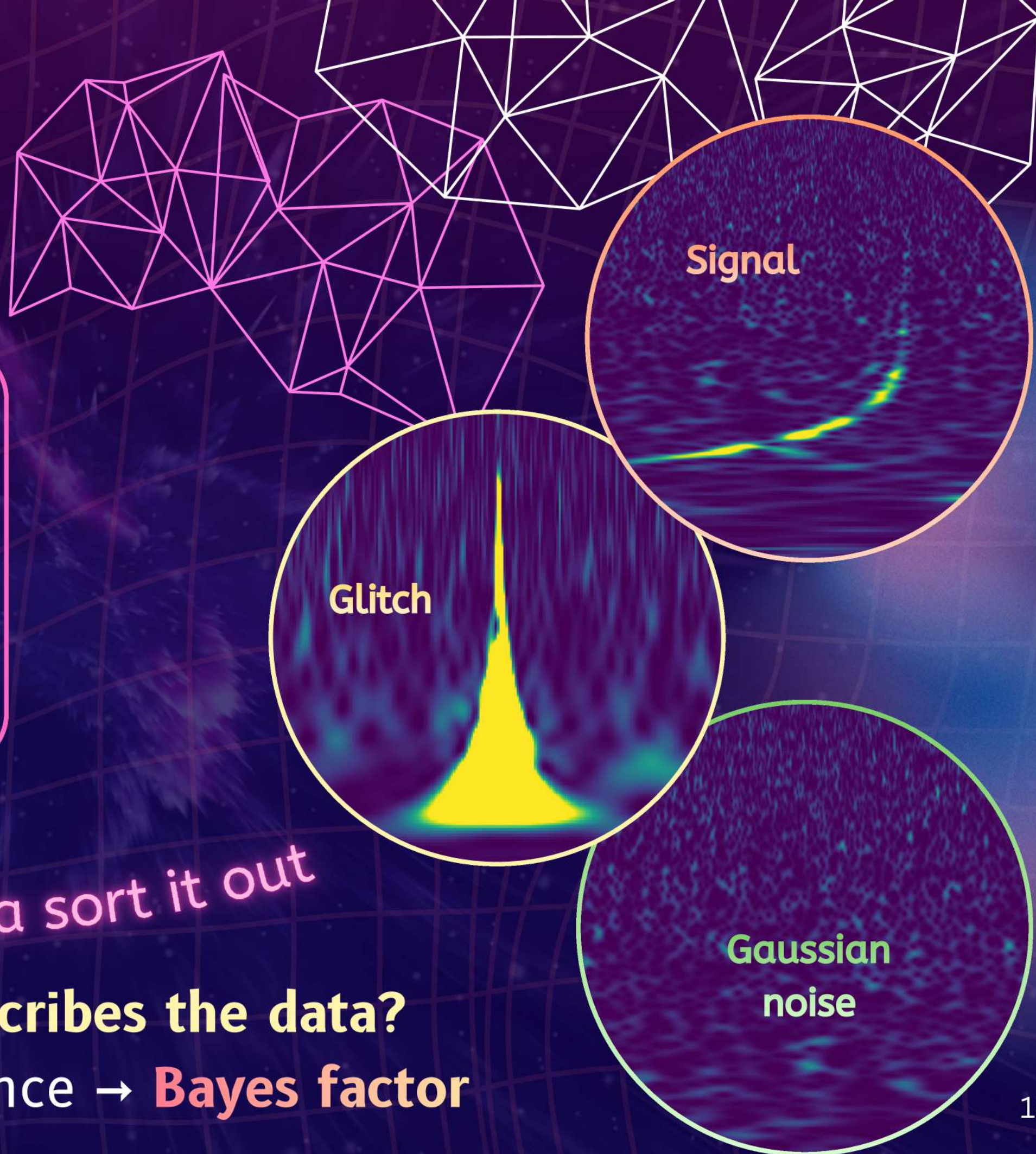
Bayes factor

Three independent models:

- Signal + Gaussian noise ($\mathcal{S}$)
- Glitch + Gaussian noise ($\mathcal{G}$)
- Gaussian noise only ($\mathcal{N}$)

Model everything and let the data sort it out

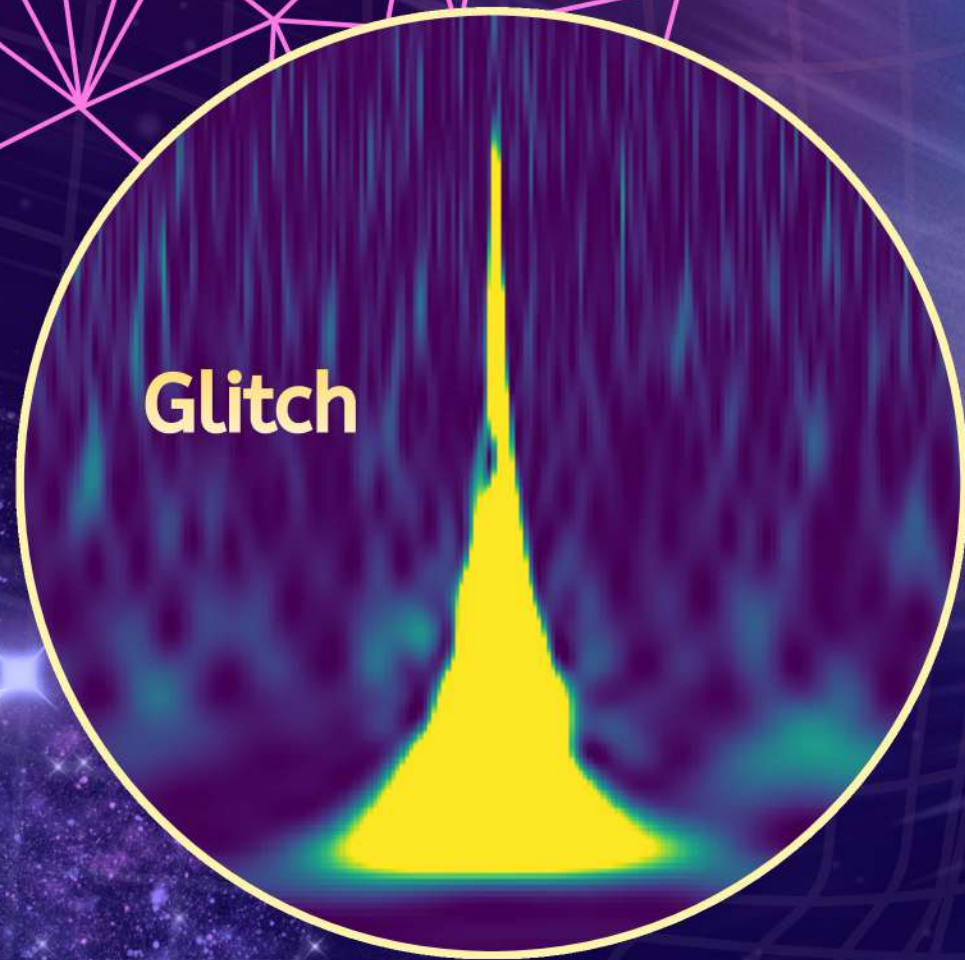
Which model best describes the data?
Compare Bayesian evidence → **Bayes factor**



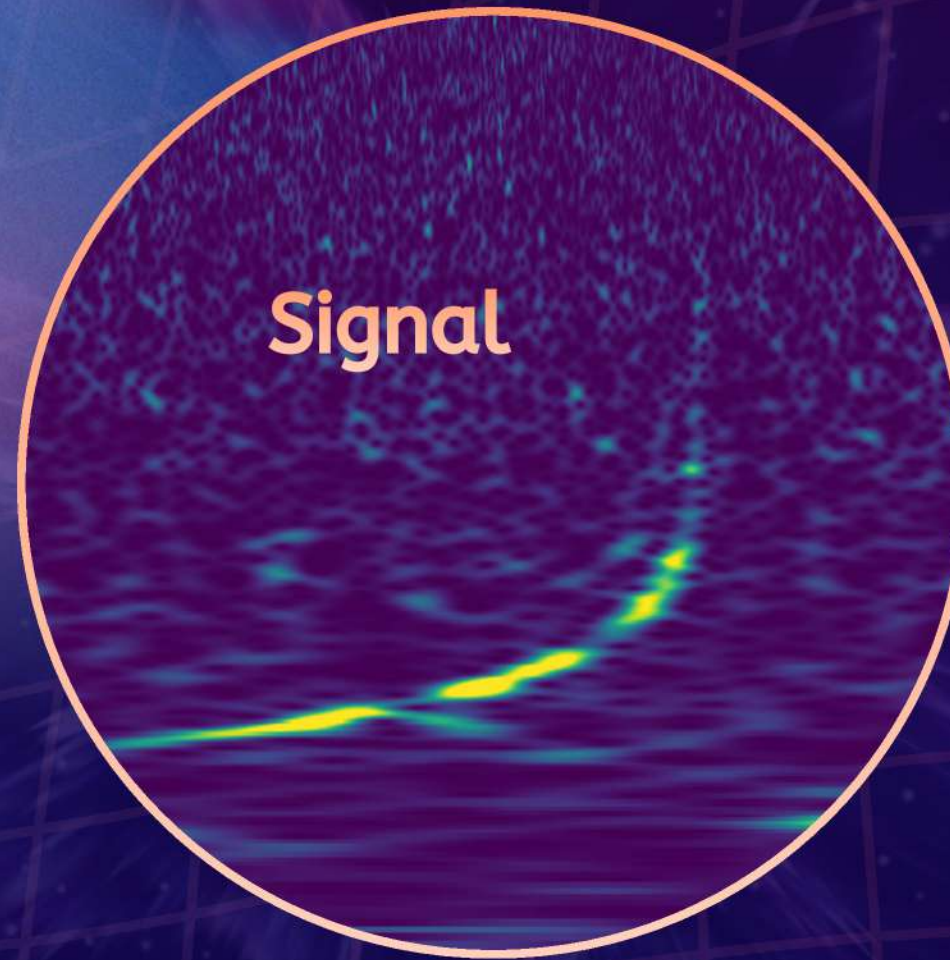
Detection statistic

Signal-versus-glitch

Bayes factor



VS



Measures *BayesWave*'s confidence in distinguishing **signals** from **glitches**
[Both non-Gaussian transients]

$$\ln \mathcal{B}_{\mathcal{S}, \mathcal{G}} = \underbrace{\ln p(\vec{s} | \mathcal{S})}_{\text{Signal evidence}} - \underbrace{\ln p(\vec{s} | \mathcal{G})}_{\text{Glitch evidence}}$$

$\ln \mathcal{B}_{\mathcal{S}, \mathcal{G}} > 0$ 
Signal > Glitch

BayesWave's role in LVK burst analyses

Spectral noise estimation

Modelling Gaussian noise floor for template-based parameter estimation of events in the **Gravitational-Wave Transient Catalogs (GWTCs)**

Waveform consistency tests

Verify the consistency between the waveform reconstructions produced by template-based analyses

Glitch subtraction

BayesWave can simultaneously model a signal and glitch, making it an effective glitch subtraction tool.

1

Multi-detector impact on *BayesWave's* detection performance



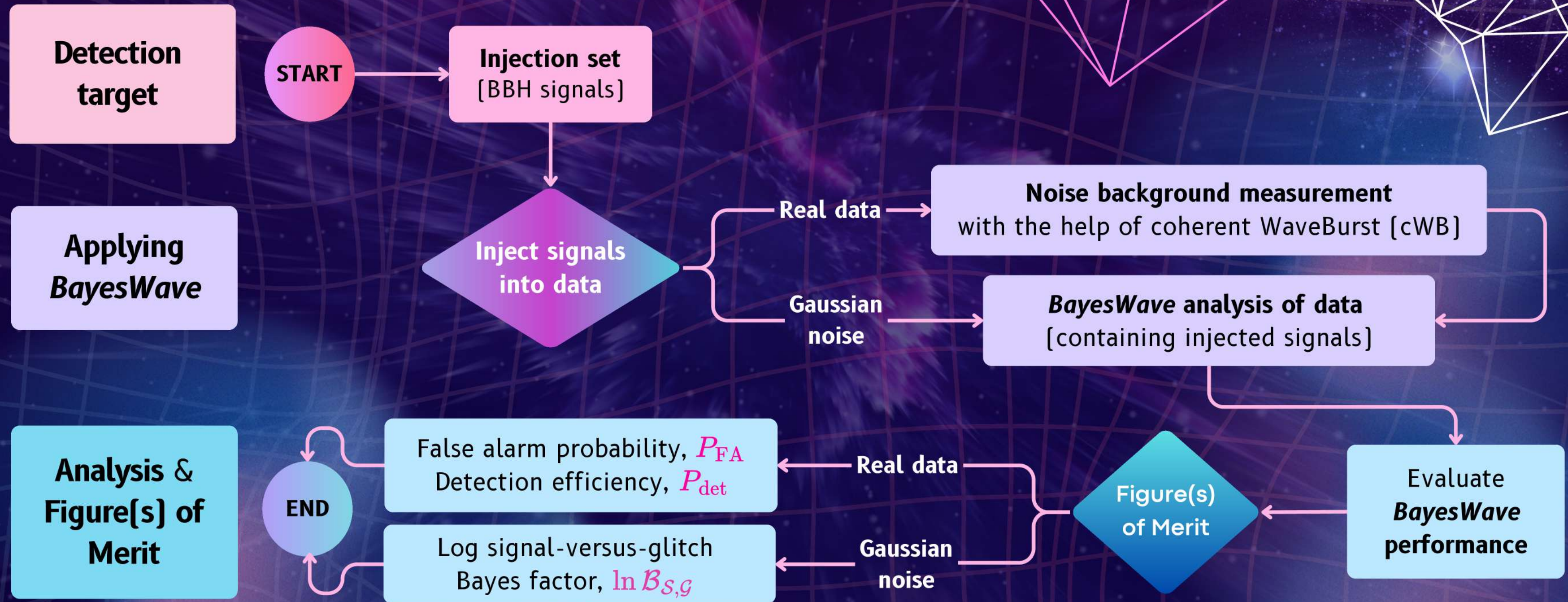
LIGO INDIA

COSMIC EXPLORER

EINSTEIN TELESCOPE

Method overview

Performance analysis





Injection set **Binary black holes**

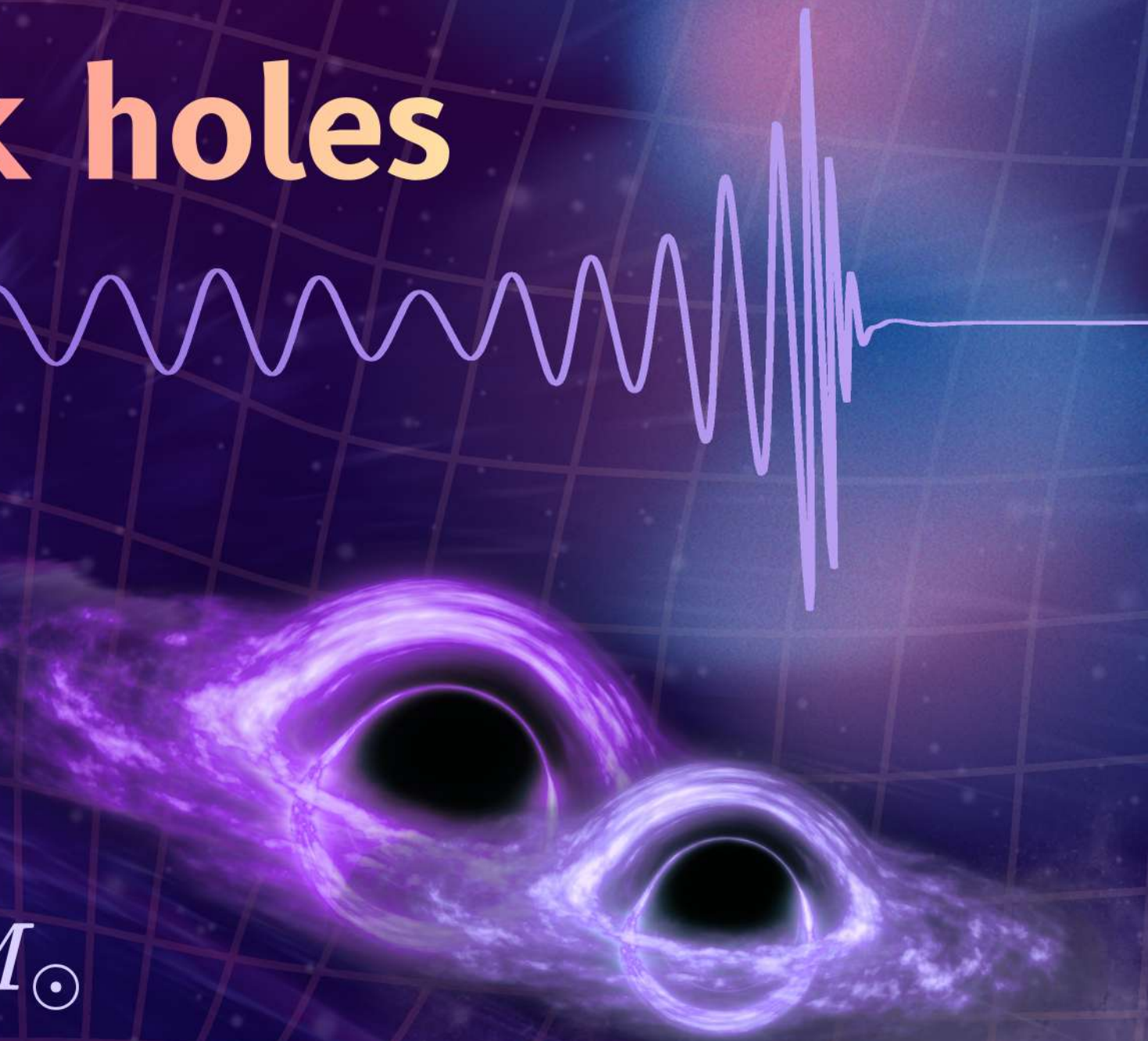
**Non-spinning
binary black holes (BBHs)**

Phenomenological waveform - IMRPhenomD

Equal component masses - $30M_{\odot}$

All sky with $\text{SNR}_{\text{net}} \sim \text{Uniform}(10, 50)$

[Network signal-to-noise ratio]



Gaussian noise injections:

Analytic predictions

$$\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} = \ln(\text{Signal evidence}) - \ln(\text{Glitch evidence})$$

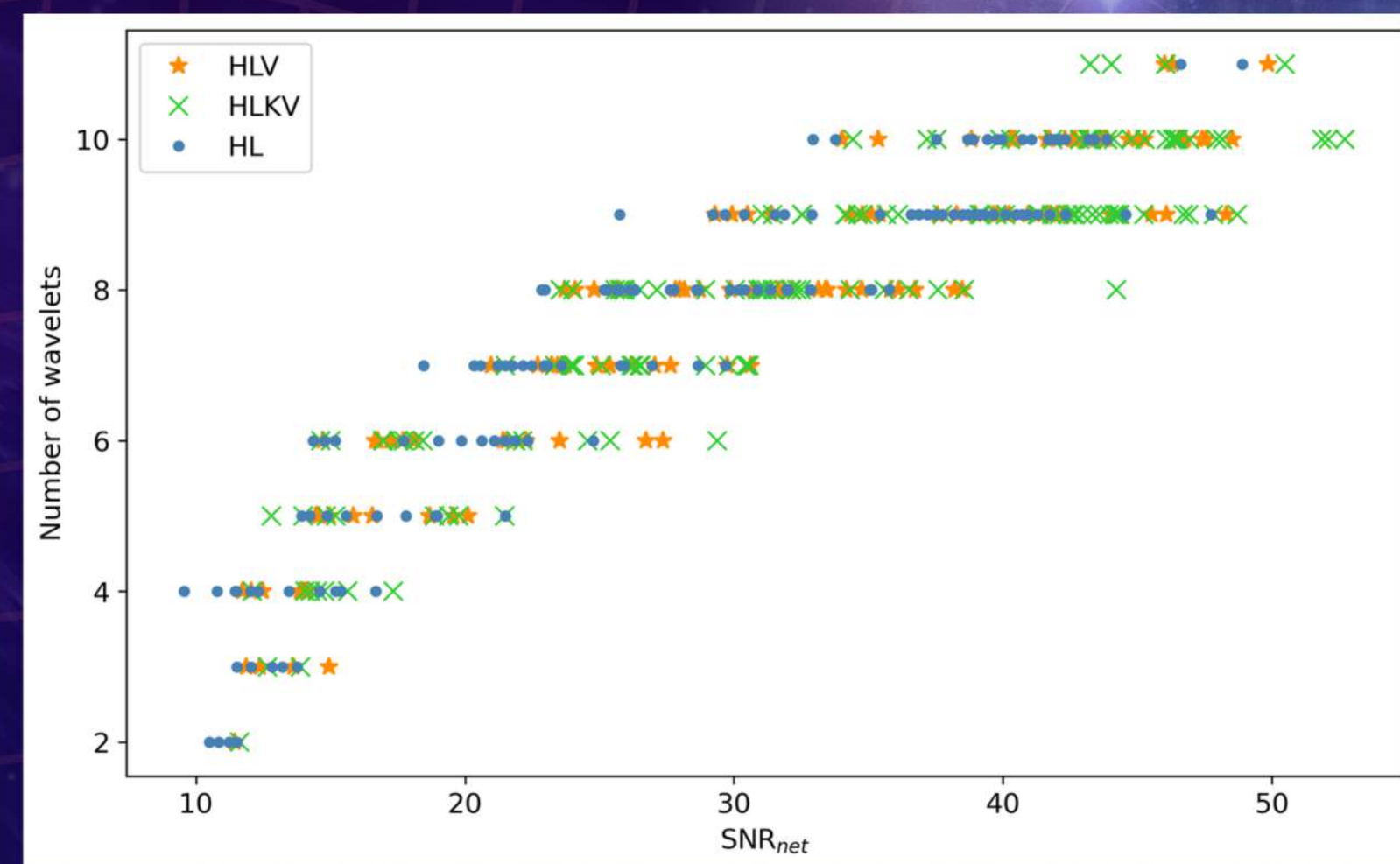
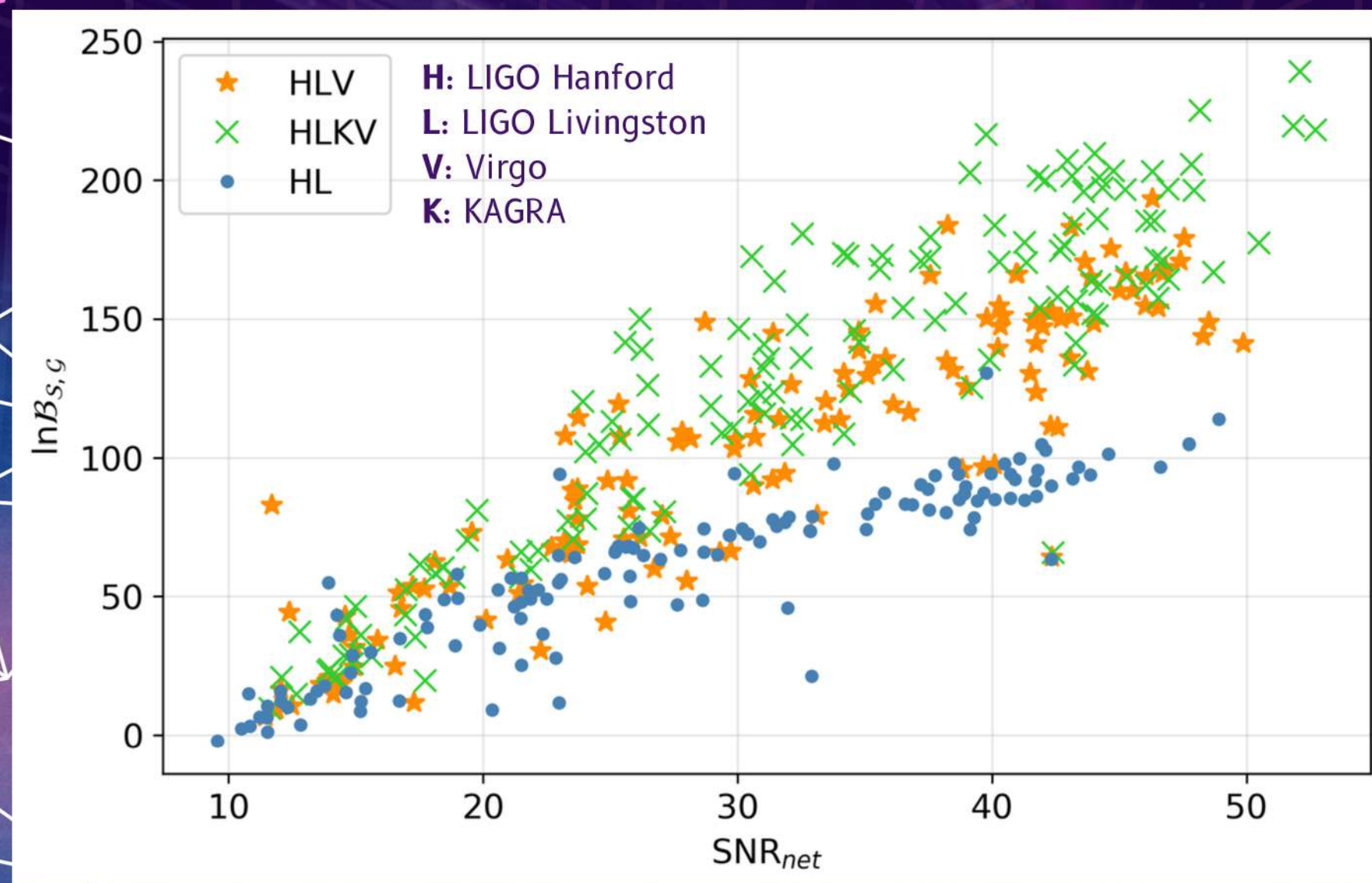
with Laplace-Fisher approximation + some assumptions...

$$\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} \sim \mathcal{I} N \ln \text{SNR}_{\text{net}}$$

Detection statistic scales with:

- number of detectors, $\mathcal{I}$
- number of wavelets, N [model complexity]
- loudness of signal, SNR_{net}

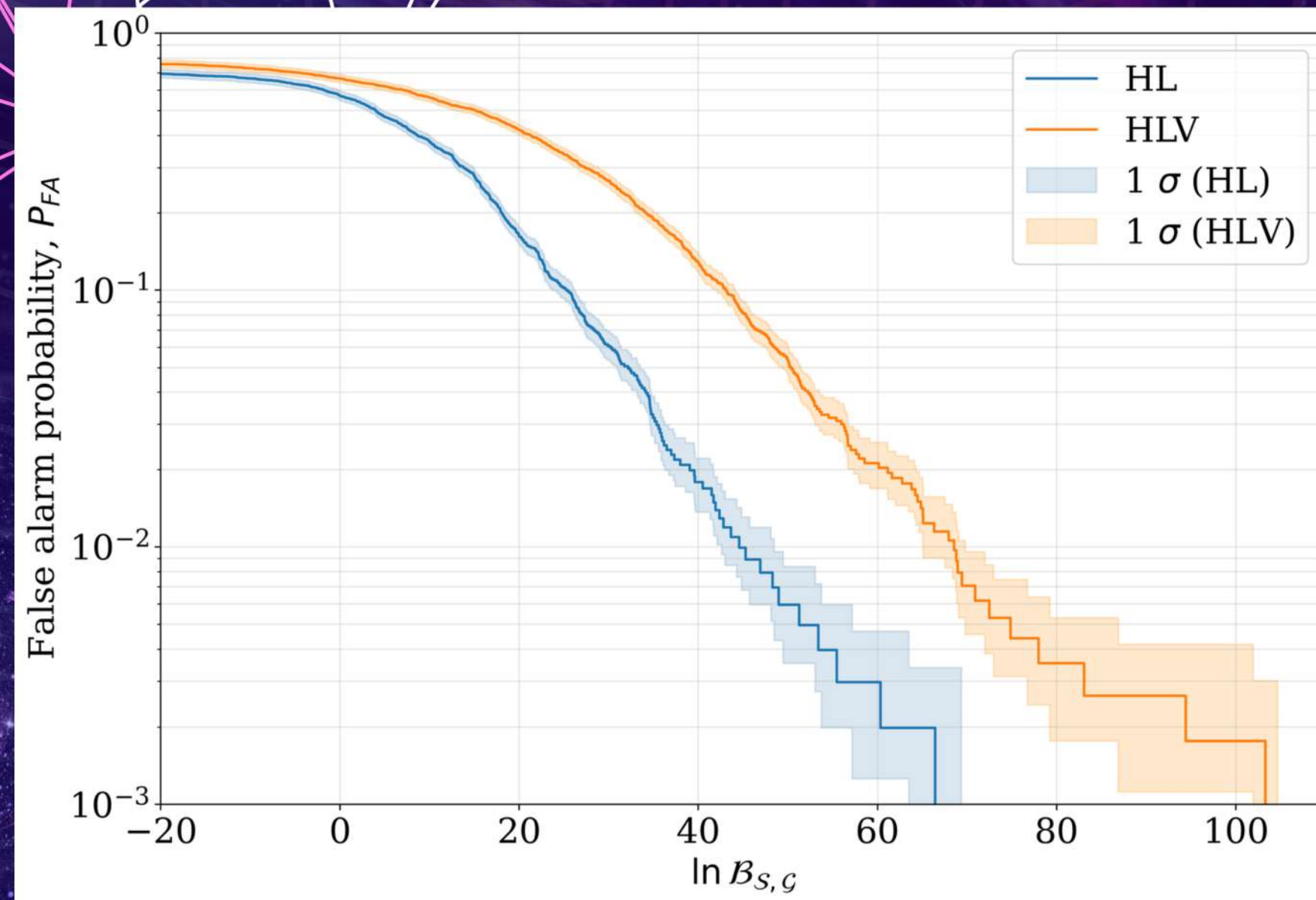
Gaussian noise injections: Empirical validation



Agrees with $\ln \mathcal{B}_{S,g} \sim \mathcal{I}N \ln \text{SNR}_{\text{net}}$

Real data (O3) injections:

Background measurements



P_{FA} : Probability of detecting **non-astrophysical triggers** [i.e. **false-alarms**] above a given $\ln \mathcal{B}_{S,G}$ threshold

coherent WaveBurst [cWB]

analyses entire background dataset

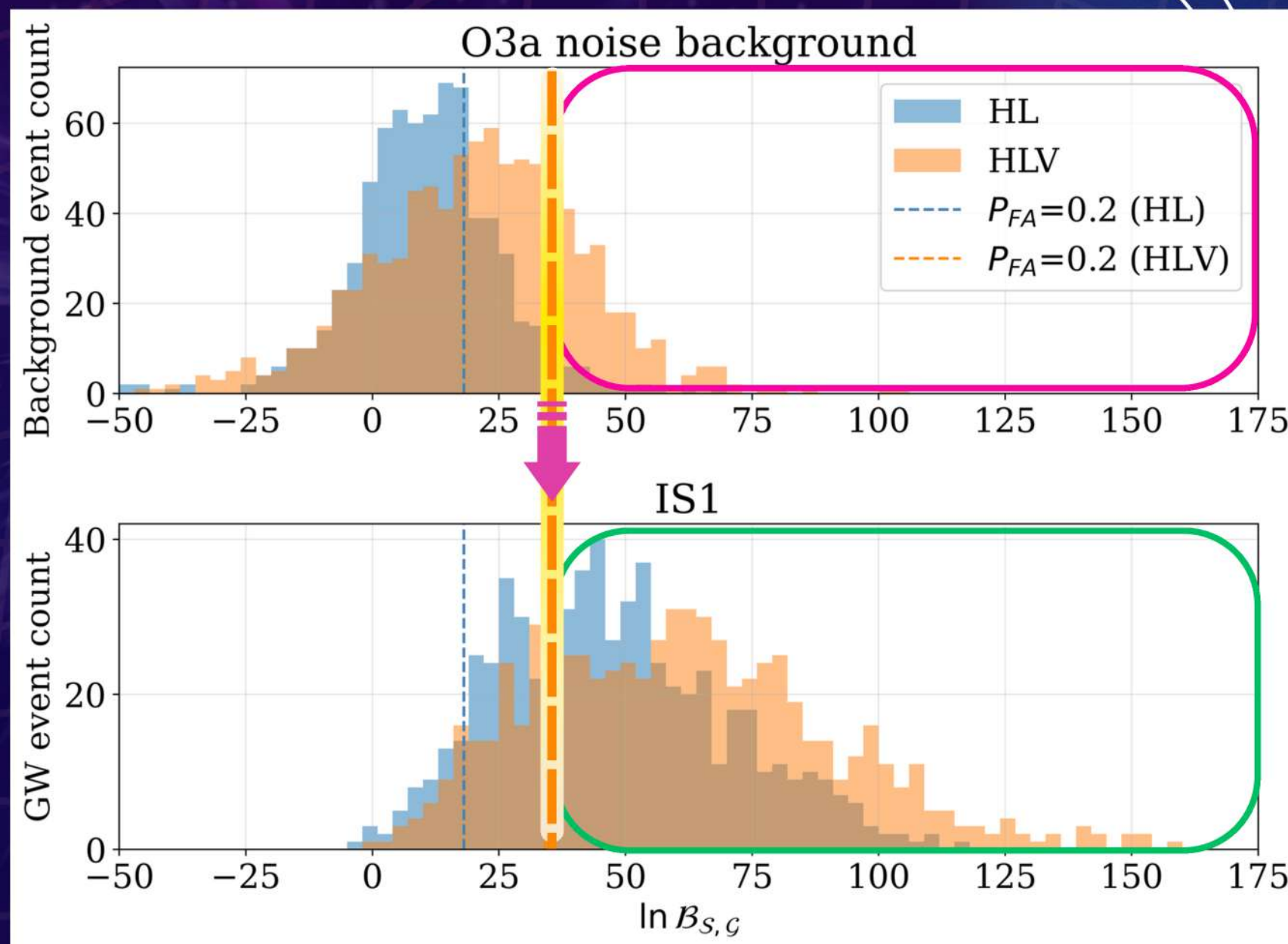
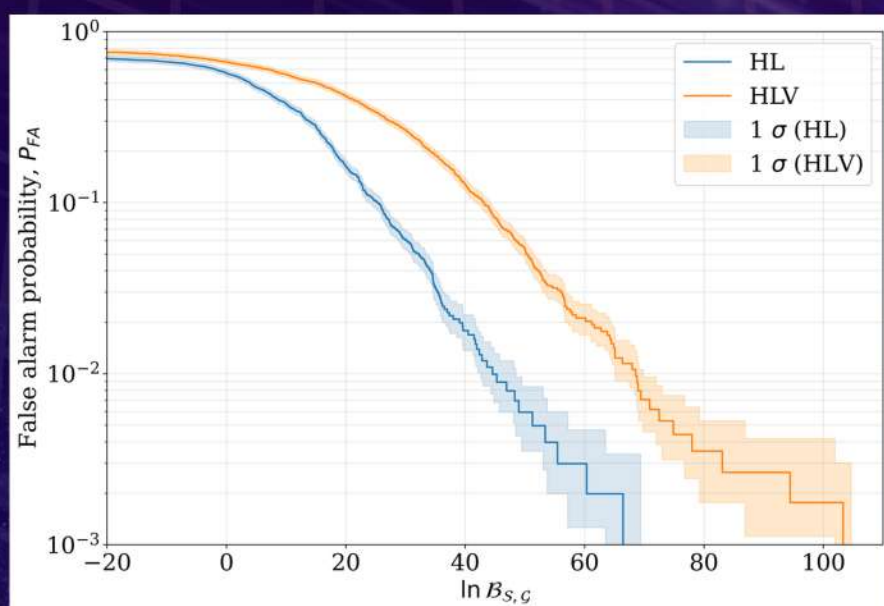
BayesWave

follows up triggers that exceeds a fixed threshold

HLV [three-detector]
is noisier than
HL [two-detector]

Real data (O3) injections:

Constructing Efficiency Curves



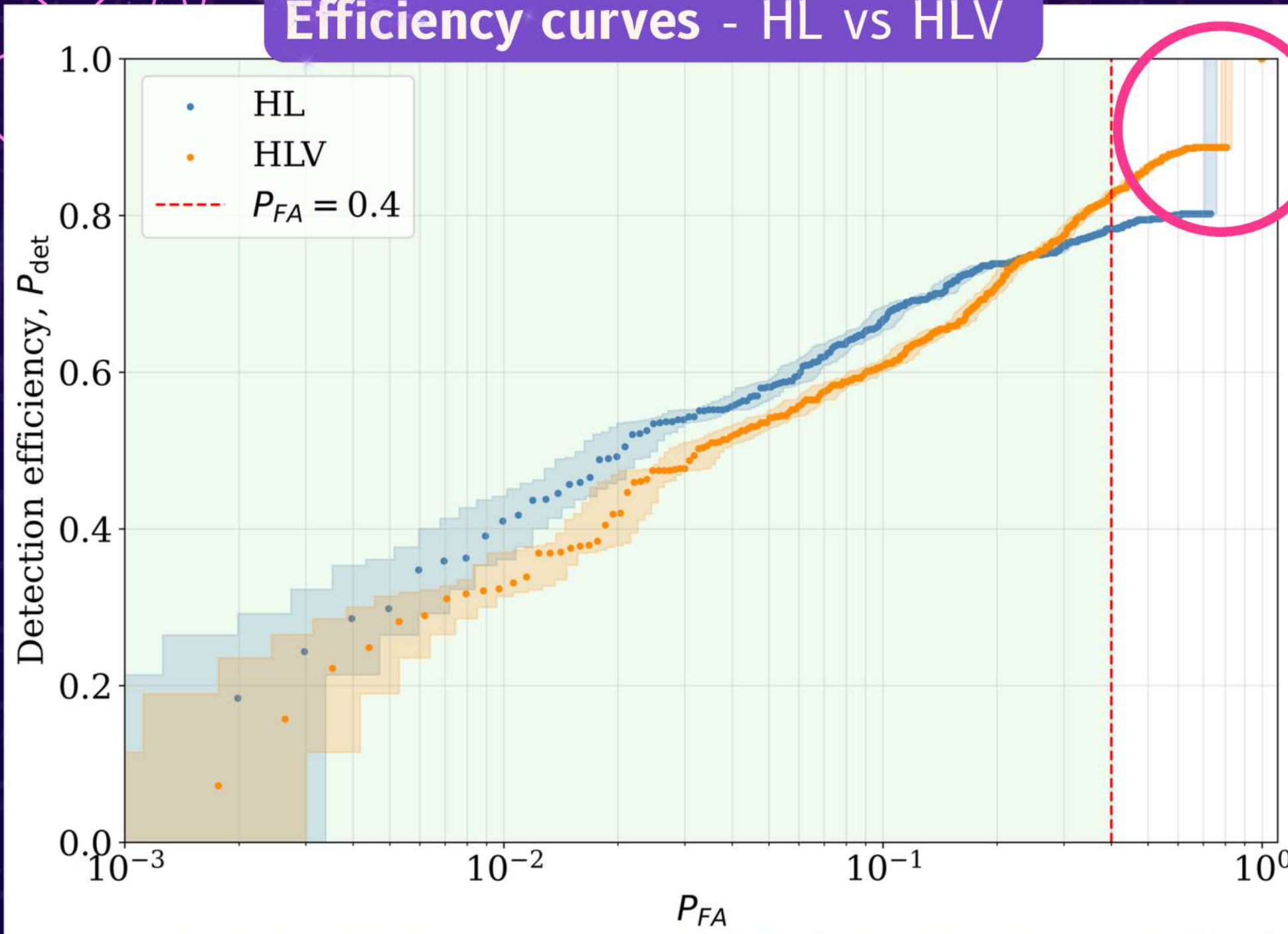
With false-alarm tolerance of $P_{FA} = 0.2$,
BayesWave's detection efficiency is $P_{det} \sim 0.7$

BBH injections



Real data (O3) injections: Detection efficiency

Efficiency curves - HL vs HLV



Gaussian-noise-like events

Assigned $P_{\text{FA}} = 1$

to indicate minimal significant

Practically meaningful regime: $P_{\text{FA}} < 0.4$

Detection efficiency of
HL [2-detector]

and

HLV [3-detector]

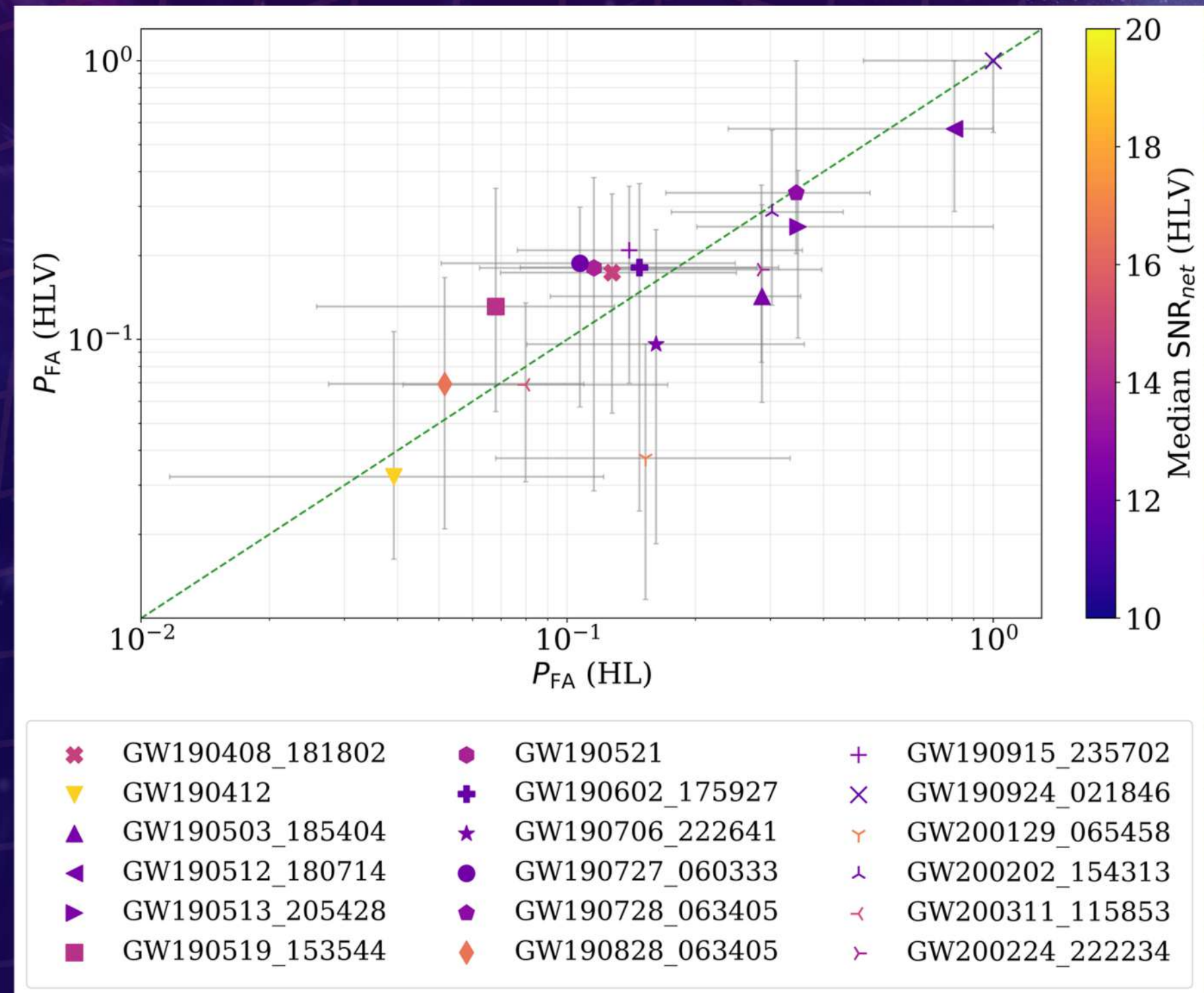
are COMPARABLE

Real data (O3) injections:

Validation with O3-like GW events

50 injections per
GW event
parameters sampled from
corresponding
match-filter posteriors

Again, comparable
 P_{FA} between
HL and HLV
for all events



Performance analysis **Summary**

BayesWave's detection statistic increases with larger detector network in Gaussian noise,

BUT...

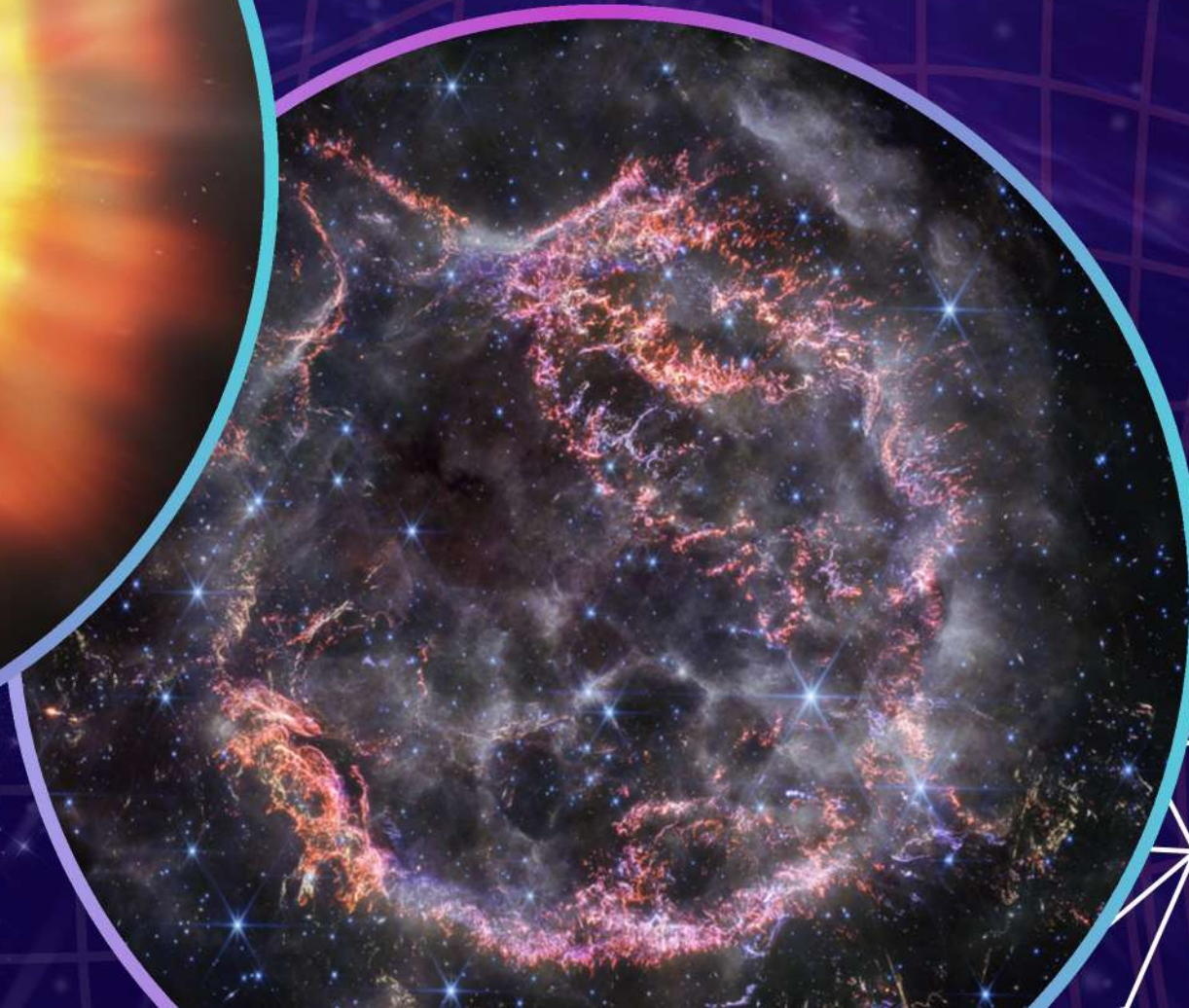
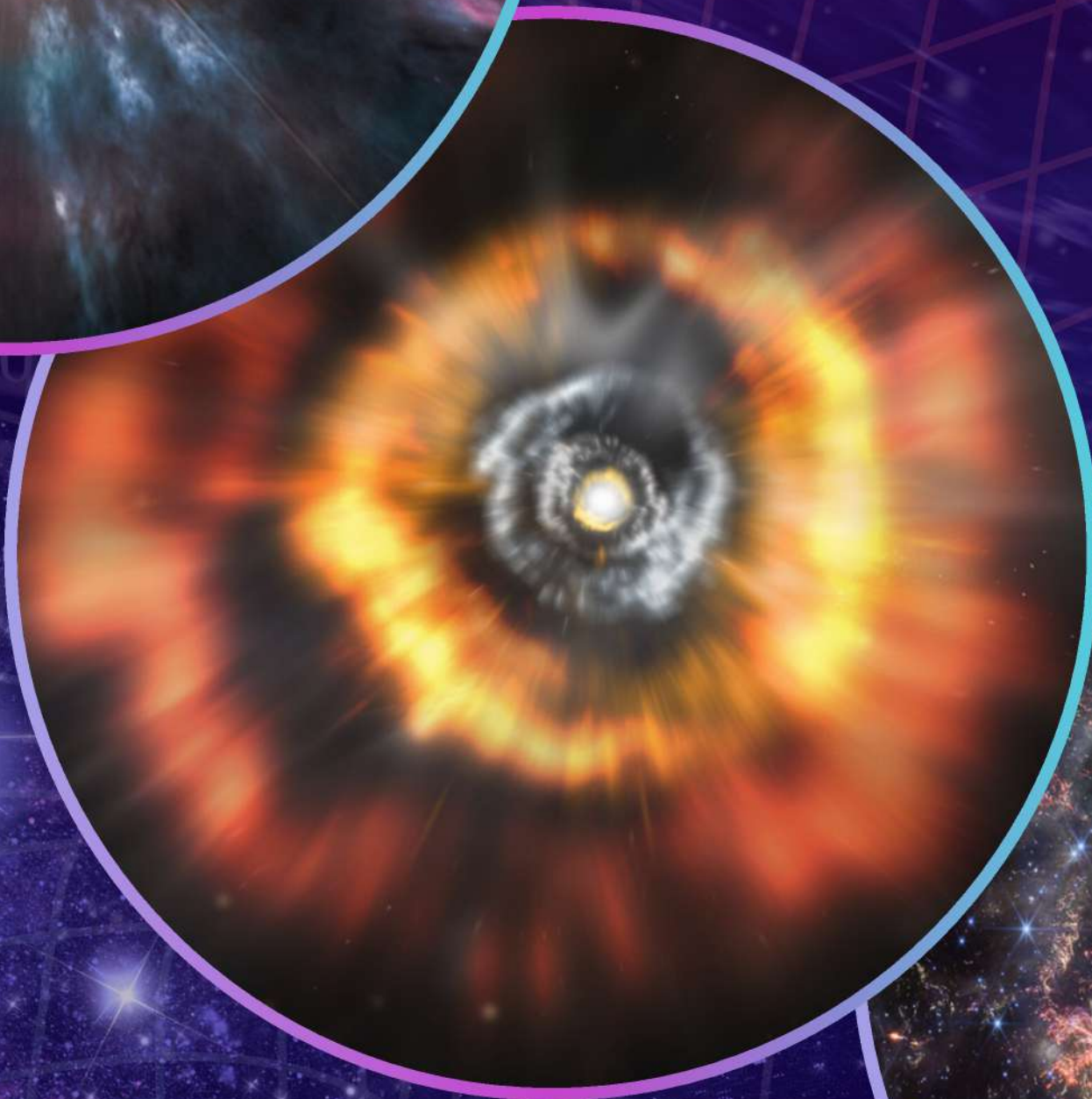
The **HL [2-detector]** and **HLV [3-detector]** networks perform comparably in the O3 observing scenario

because more frequent glitches in larger detector networks offsets the advantage of higher detection statistics

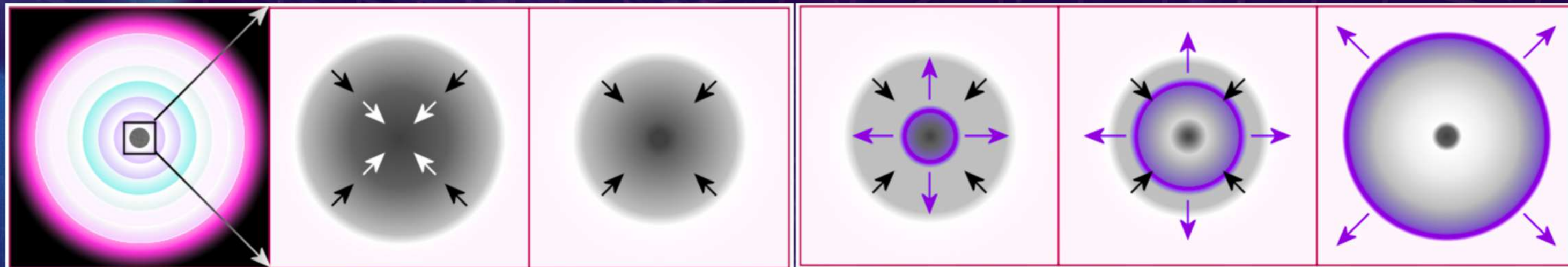
2

Dedicated-frequency
analysis of

**Core-collapse
supernovae**



Introduction to Core-collapse supernovae (CCSNe)



- Stars with mass $> 8M_{\odot}$
- Collapsing outer layer rebound off proto-neutron star (PNS) core
- Shock waves break out >>> high-energy explosion

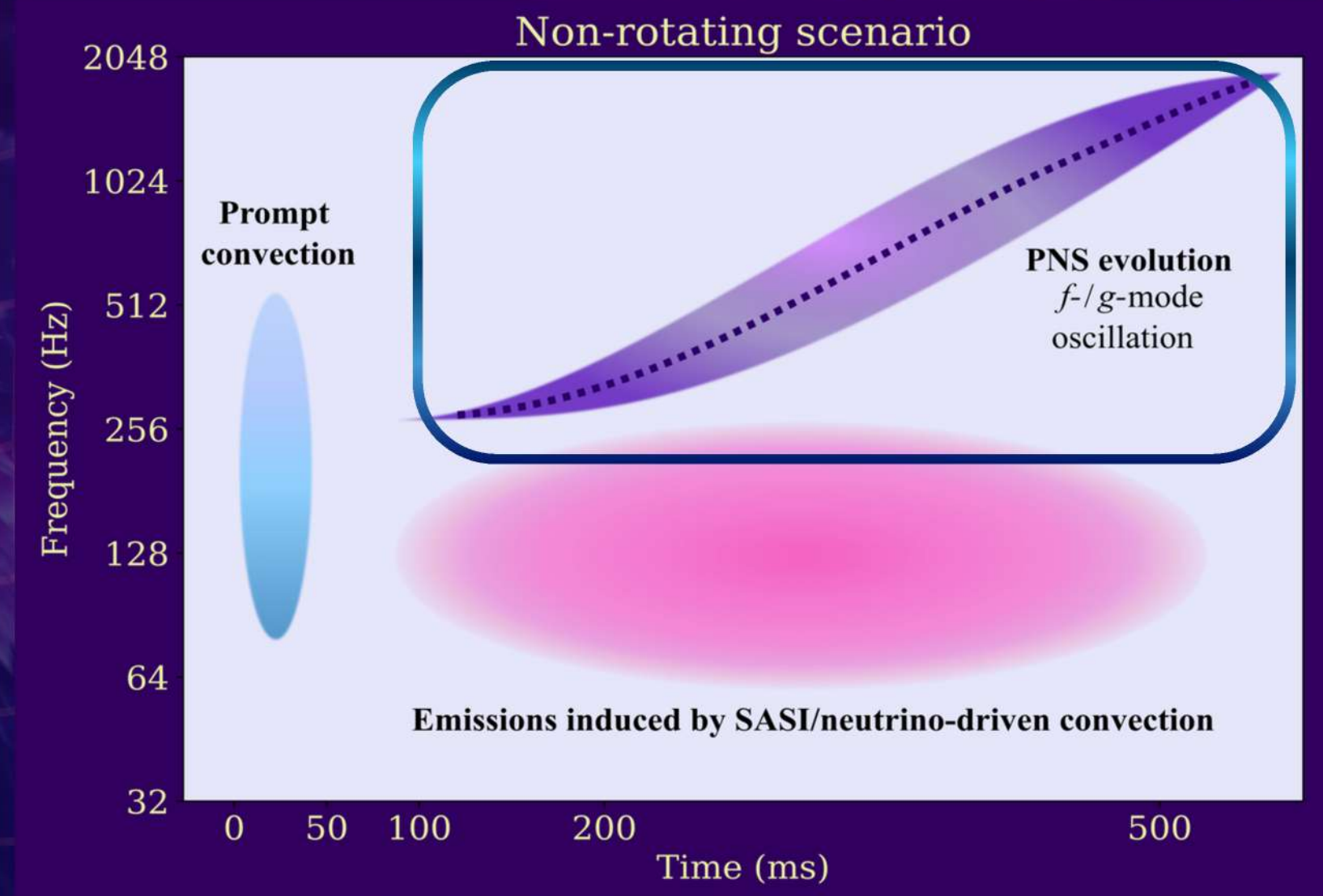
Shock revival mechanism not fully understood.
Need information from stellar interior.

Frequency signatures of GWs from CCSNe

High-frequency [$\sim 400 - 2000$ Hz]

Protoneutron-star [PNS] oscillations

Constrains PNS properties (e.g. mass/radius)
and Equation of state,
but does not imply explosion



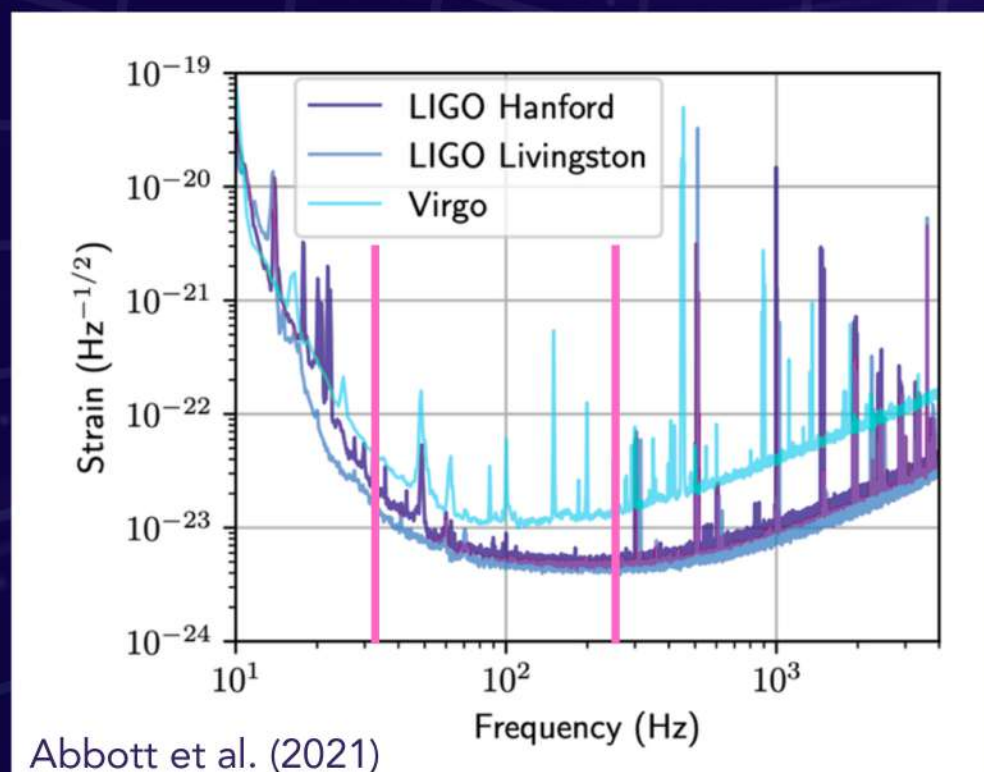
Frequency signatures of GWs from CCSNe

Low-frequency ($\lesssim 250$ Hz)

Hydrodynamic instabilities

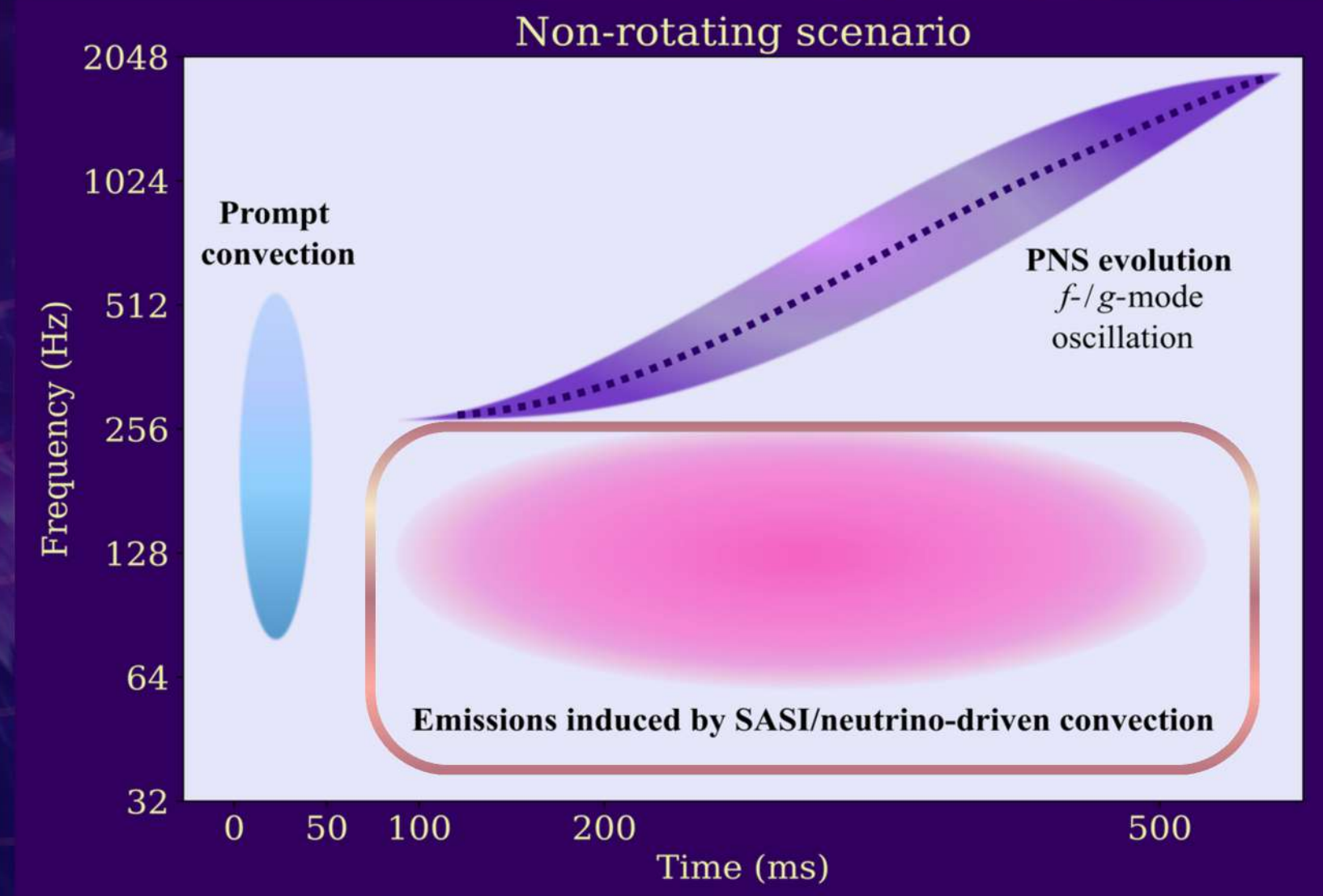
Standing accretion shock instability [SASI]

Neutrino-driven convection

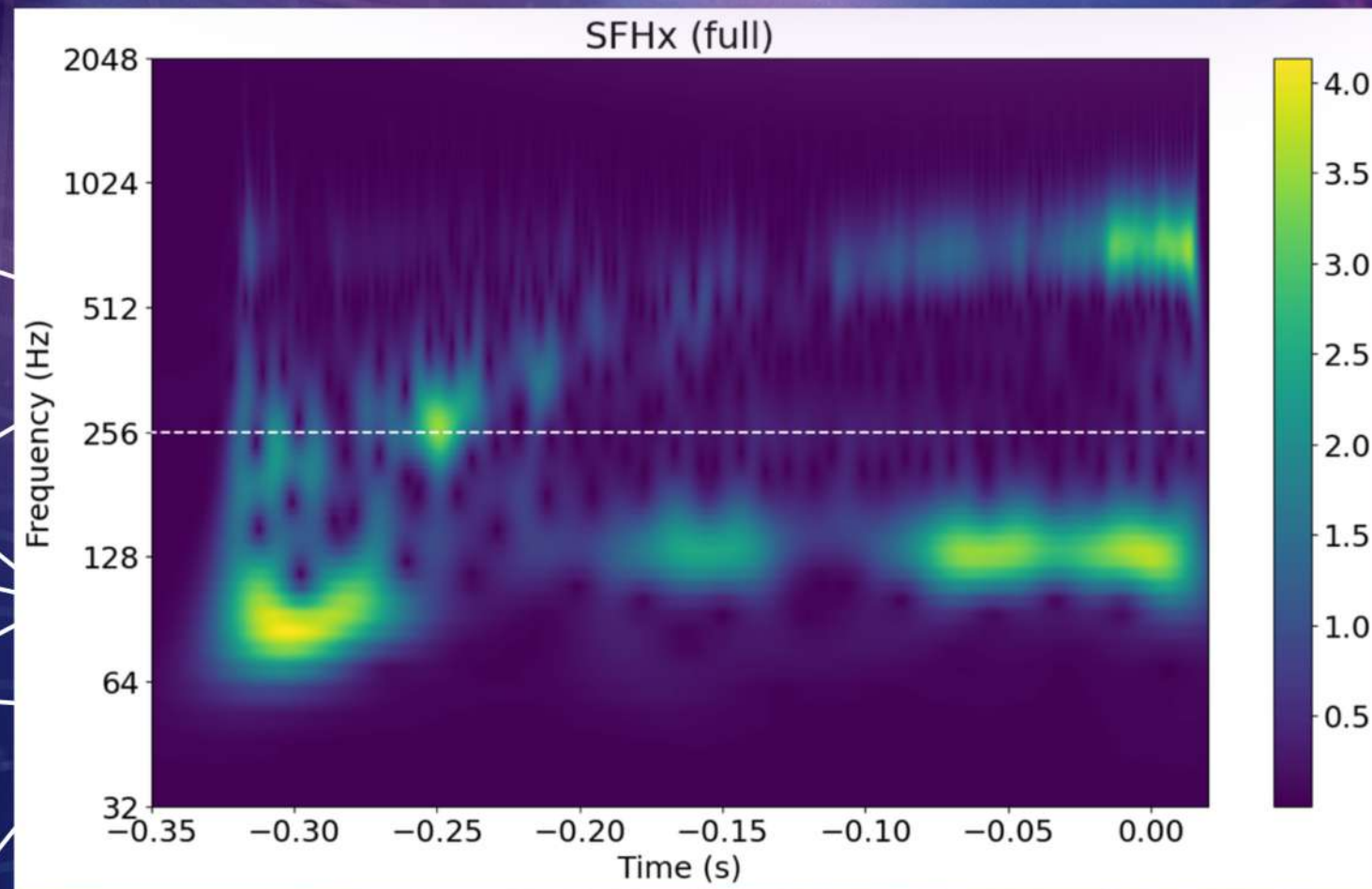


potential explosion
drivers ✓

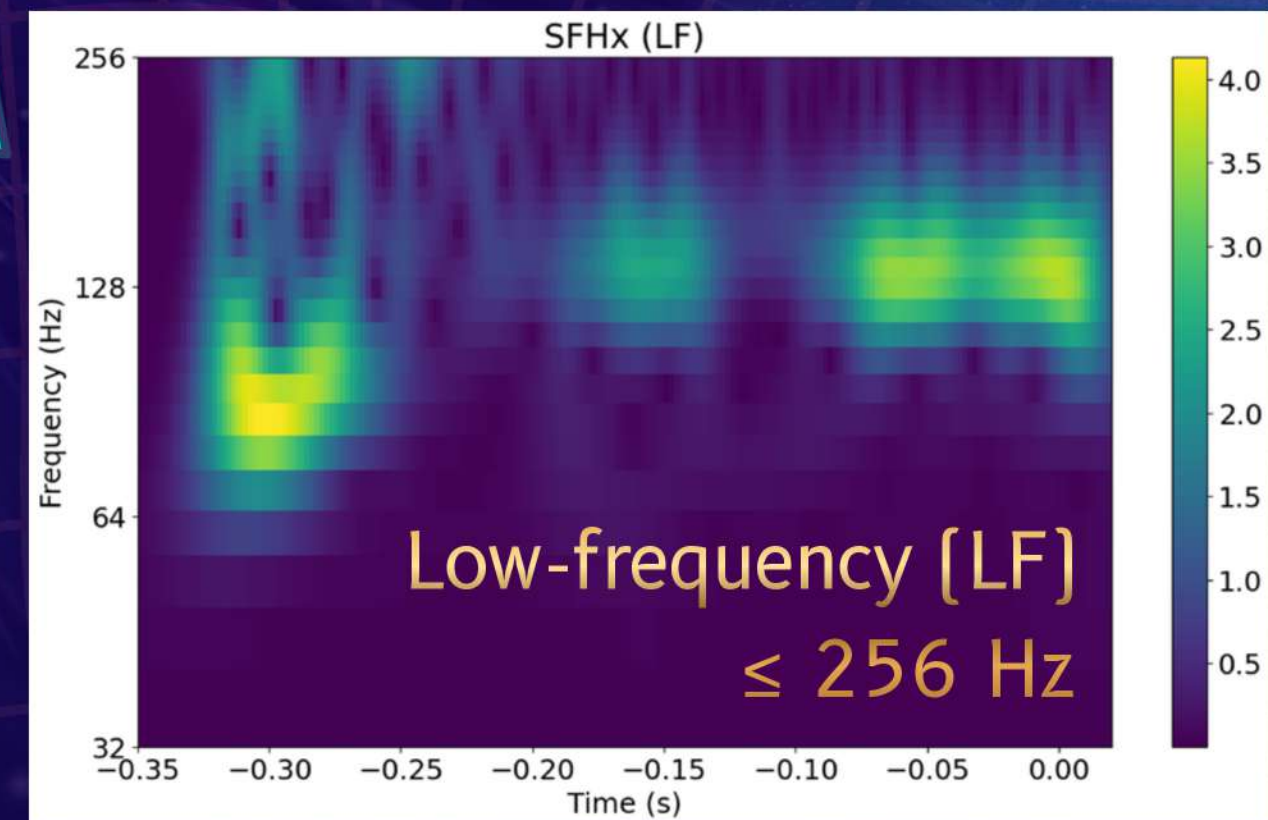
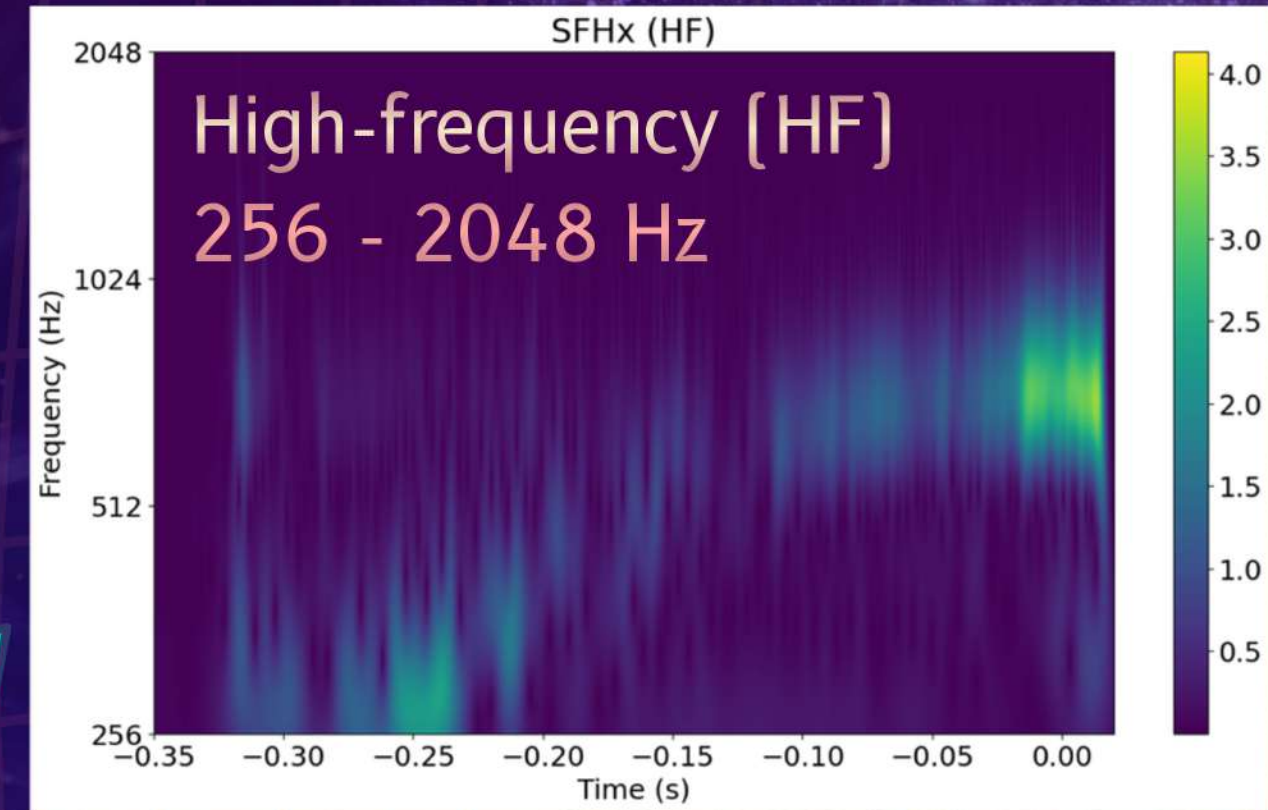
falls within the most
sensitive band of
LIGO/Virgo ✓



Dedicated-frequency analysis



Full band : 32 - 2048 Hz



IS IT POSSIBLE TO...

Enhance sensitivity to frequency-specific GW signatures by ignoring contributions from irrelevant frequency ranges?



Constrain CCSN explosion models through the detection of low-frequency GW signatures?

Analysis dataset

CCSNe waveforms



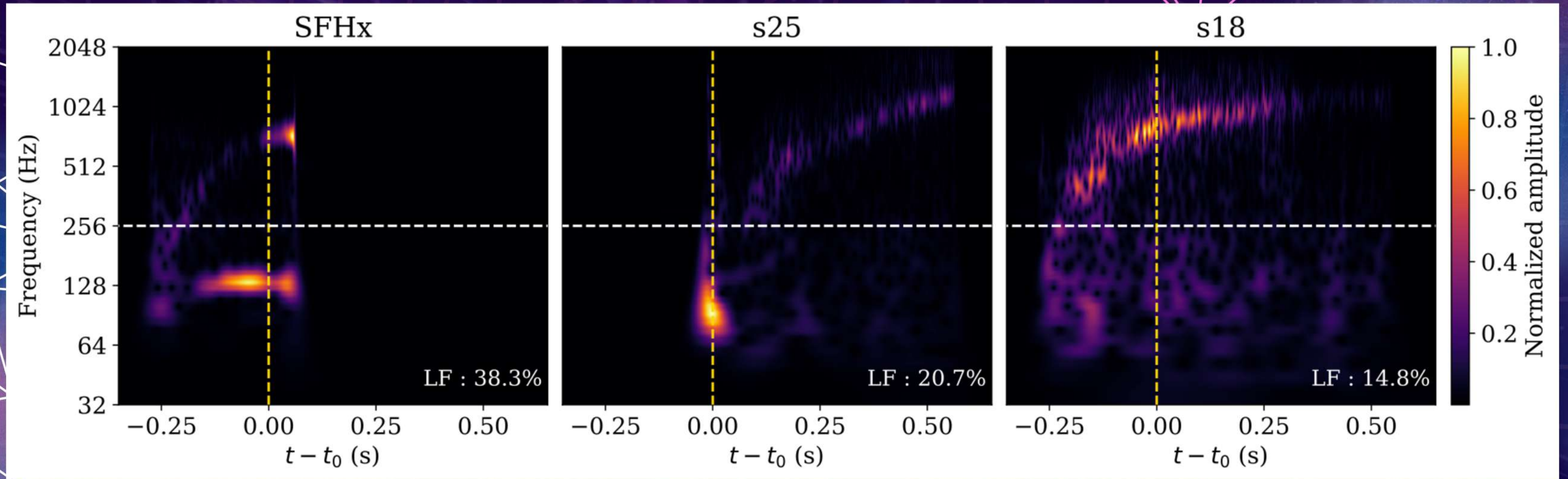
	Progenitor mass ($M_{\odot}$)	SASI / neutrino-driven convection?	Average LF power (%)
SFHx (Kur+16)	15	Yes	36.6
s25 (Rad+19)	25	Yes	19.4
D15-3D (Mez+20)	15	Yes	18.4
mesa20_pert (Oco+18)	20	Yes	16.2
s18 (Pow+18)	18	No	14.9

150 waveforms per
CCSN model

Injected into **O3 data**
of the **HL [2-detector]**
network

CCSNe waveforms

Spectrograms



Weaker SASI/convection,
decreasing LF power

Method overview

Hierarchical analysis

coherent WaveBurst [cWB]

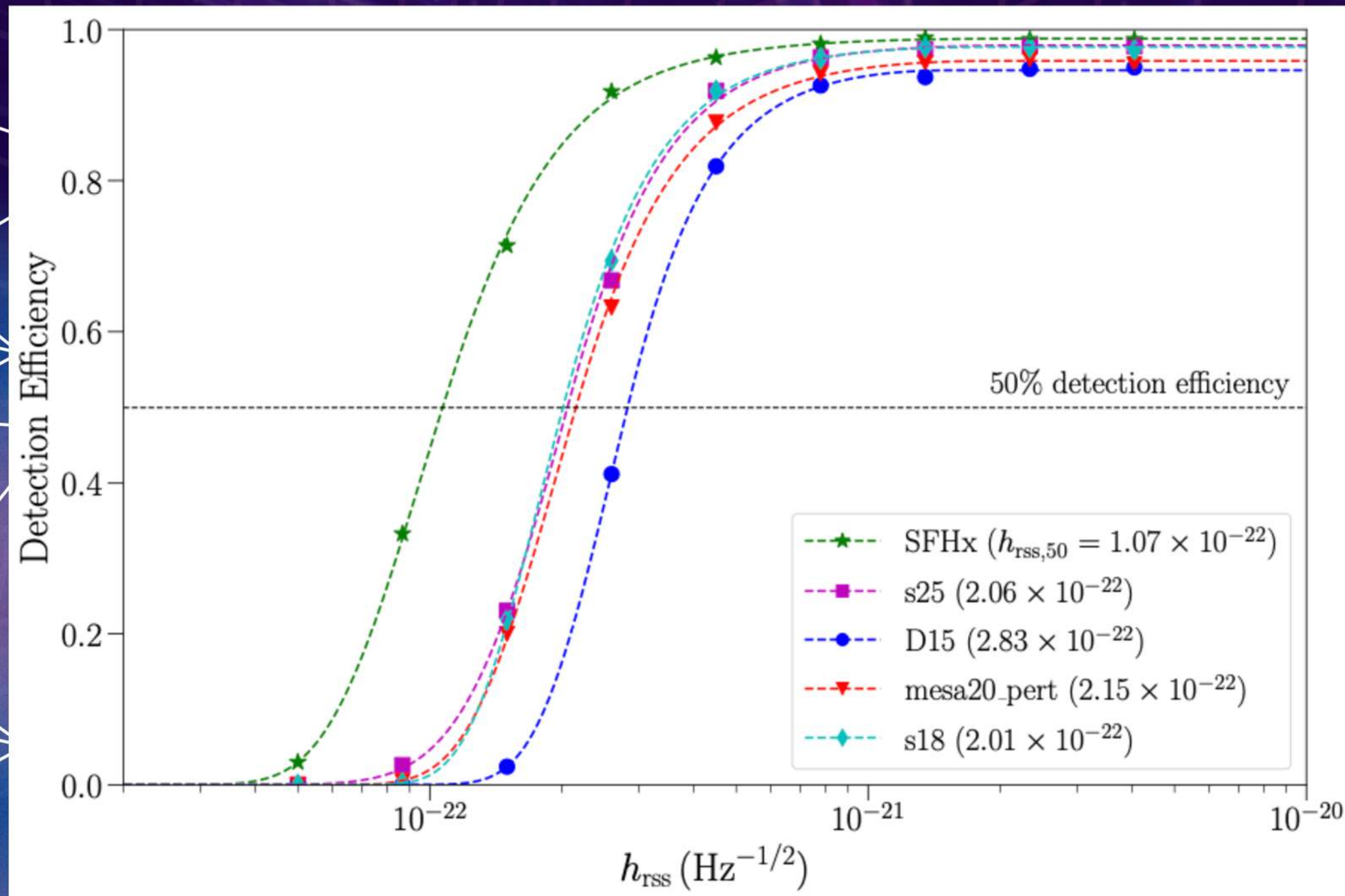
- Fast and computationally efficient
- Suitable for analyzing large datasets
- **In this work:**
 - To identify eligible CCSNe candidates for dedicated-frequency follow-up

BayesWave

- Computationally expensive due to transdimensional sampling
- Serves as a follow-up tool in practice
- **In this work:**
 - Follow-up cWB CCSNe candidates in the full band **AND** HF/LF bands

CCSNe waveforms

Injection amplitudes



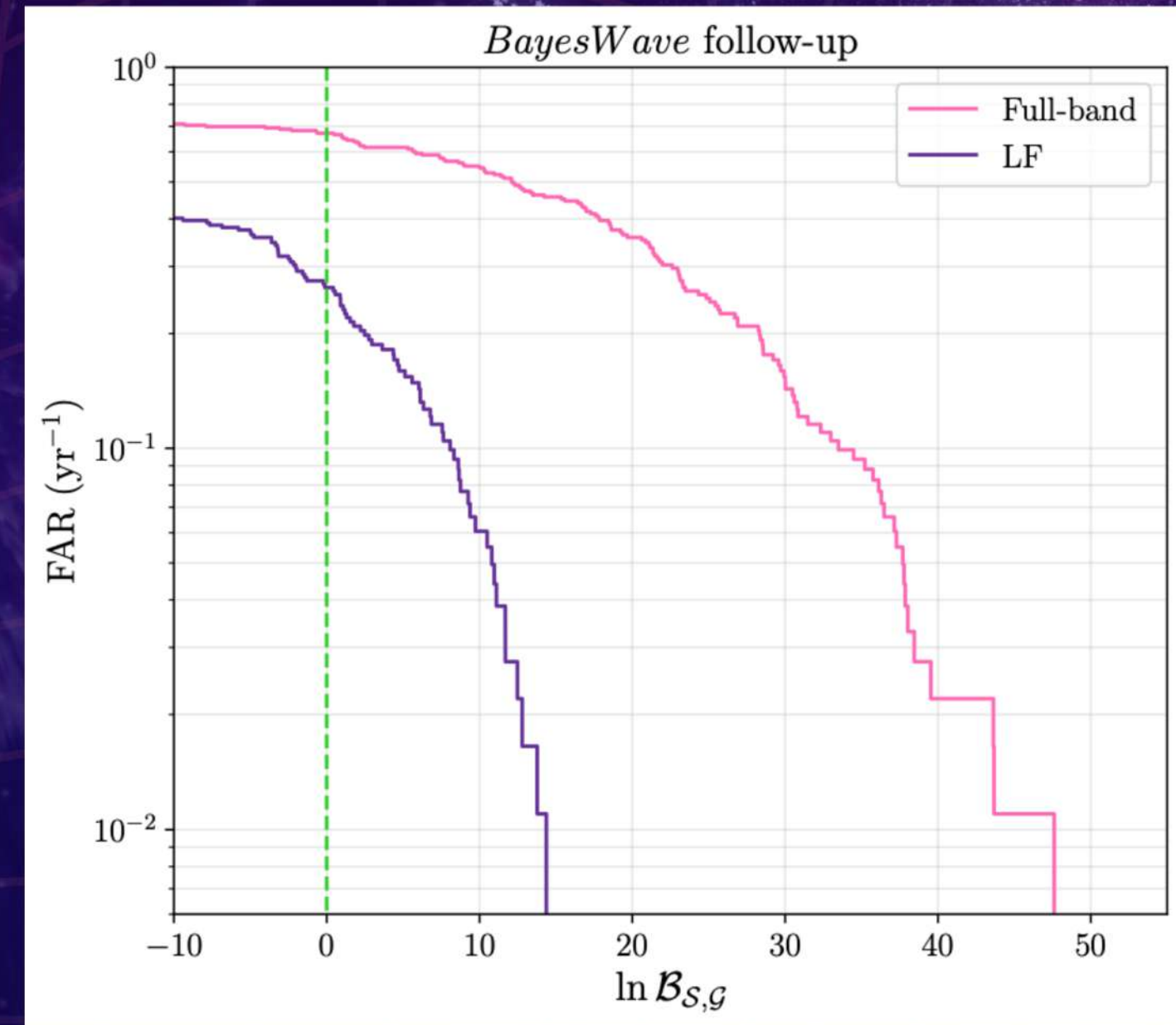
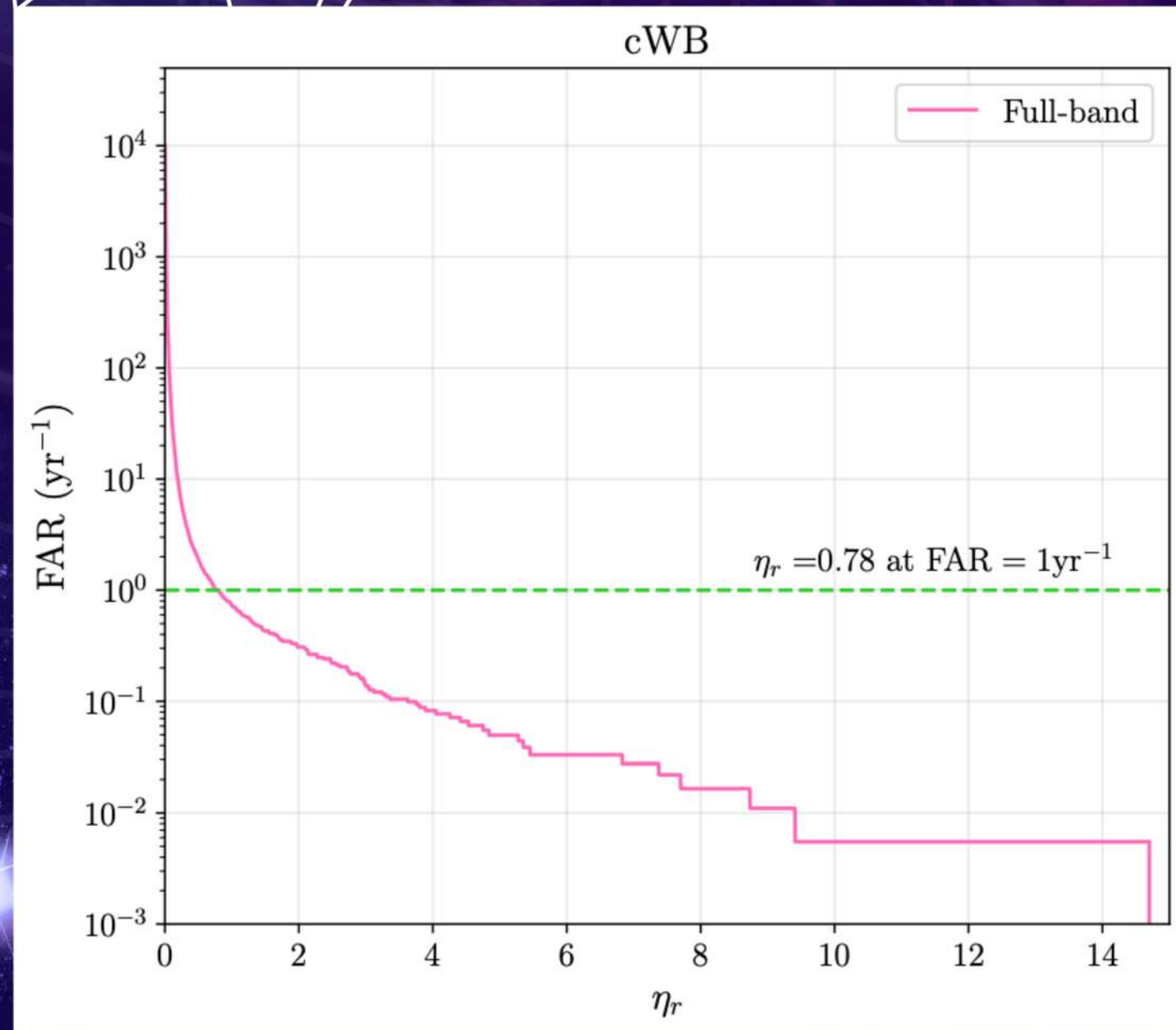
$h_{\text{rss},50}$

Amplitude at which cWB detects the CCSN with 50% detection efficiency, with false-alarm rate [FAR] below 1 yr^{-1}

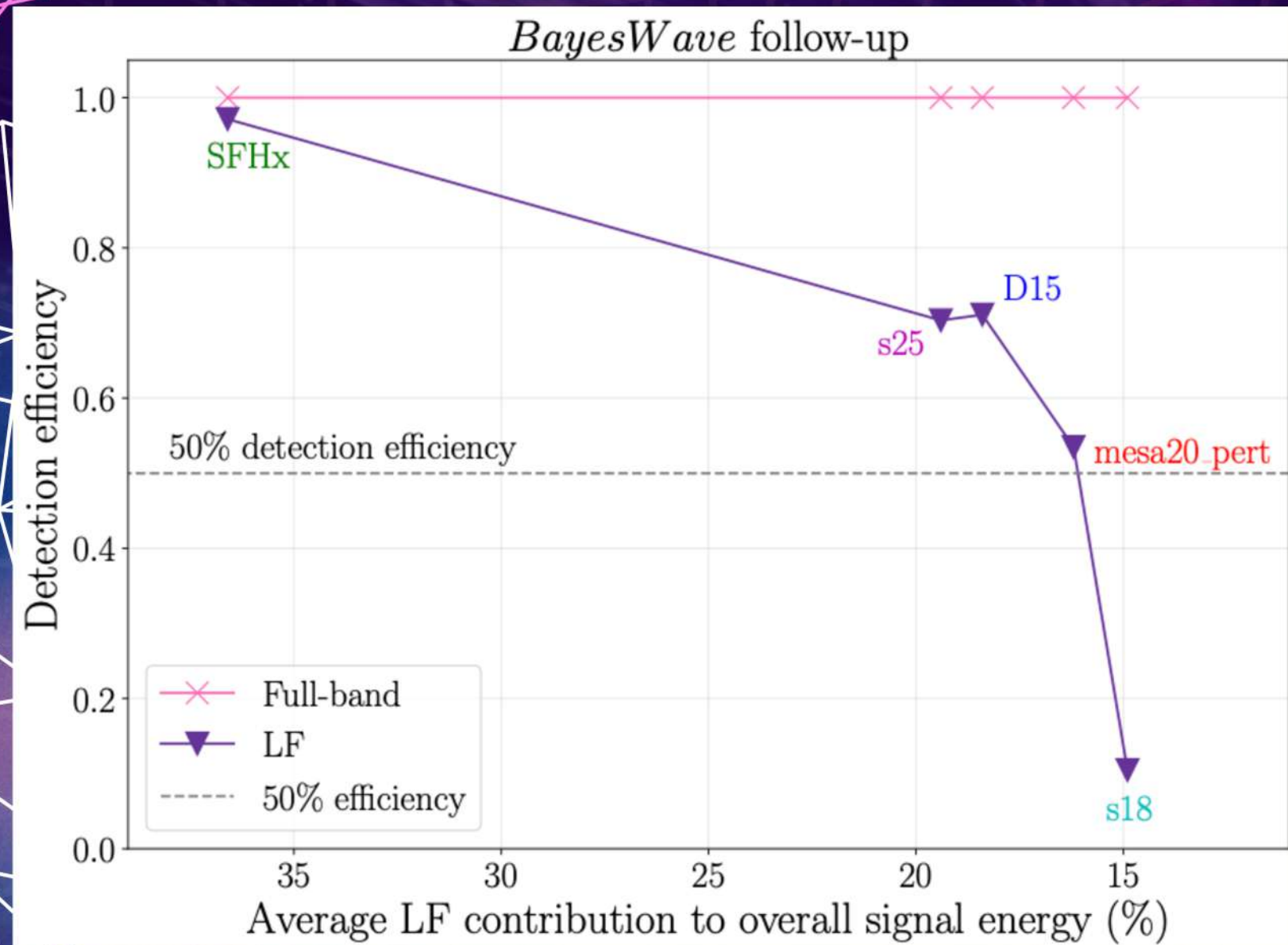
Model	E_{GW} [$M_{\odot} c^2$]	f_{peak} [Hz]	$h_{\text{rss},50}$ [$\text{Hz}^{-1/2}$]	r_{50} [kpc]
SFHx	1.1×10^{-9}	267	1.07×10^{-22}	7.8
s25	2.7×10^{-8}	1132	2.06×10^{-22}	4.9
D15	8.9×10^{-9}	1102	2.83×10^{-22}	2.1
mesa20_pert	9.4×10^{-10}	1103	2.15×10^{-22}	0.9
s18	1.6×10^{-8}	818	2.01×10^{-22}	5.3

Real data (O3) injections

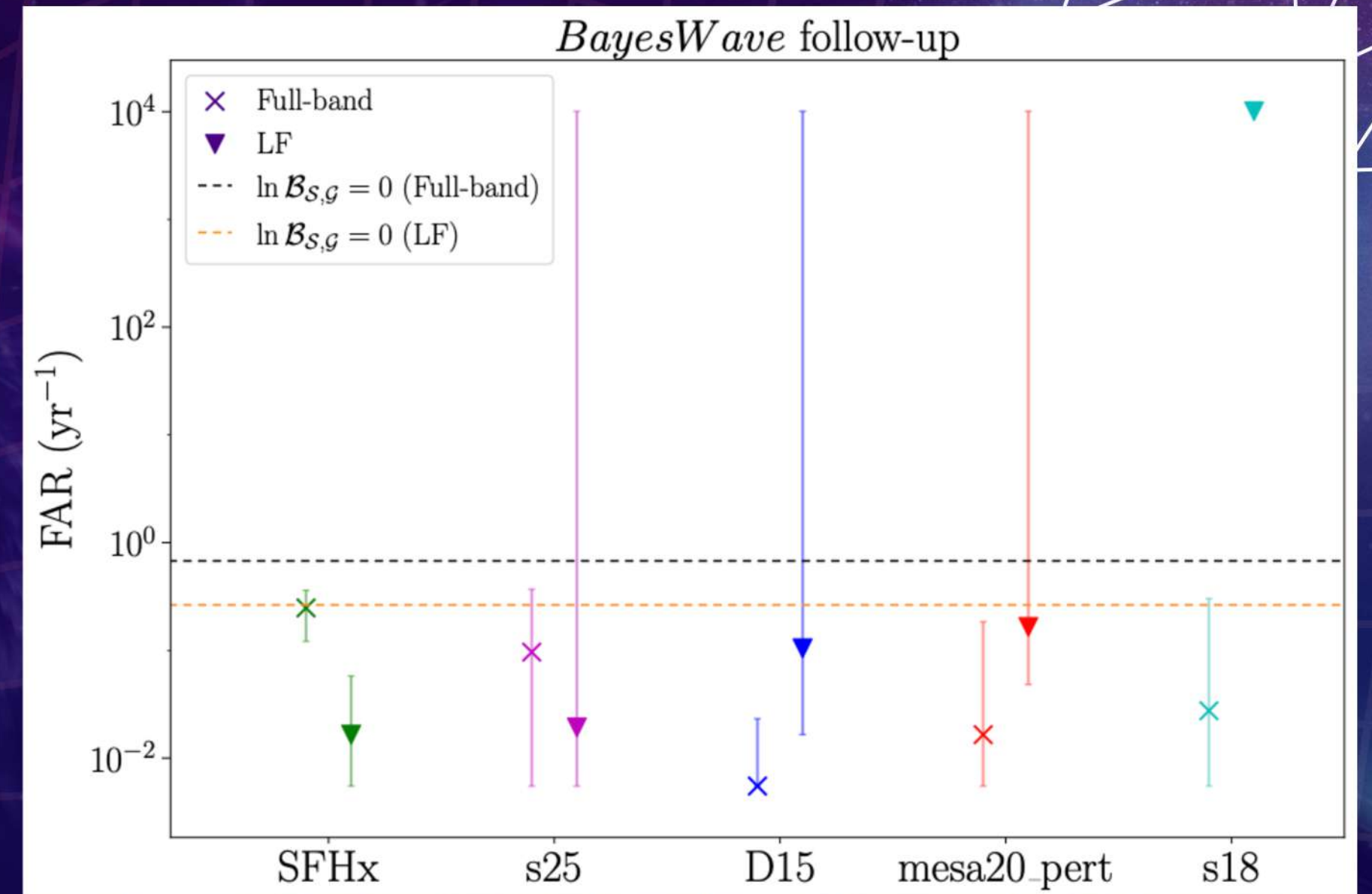
Background measurements



Detectability of Low-frequency GW signatures



Decreasing LF power

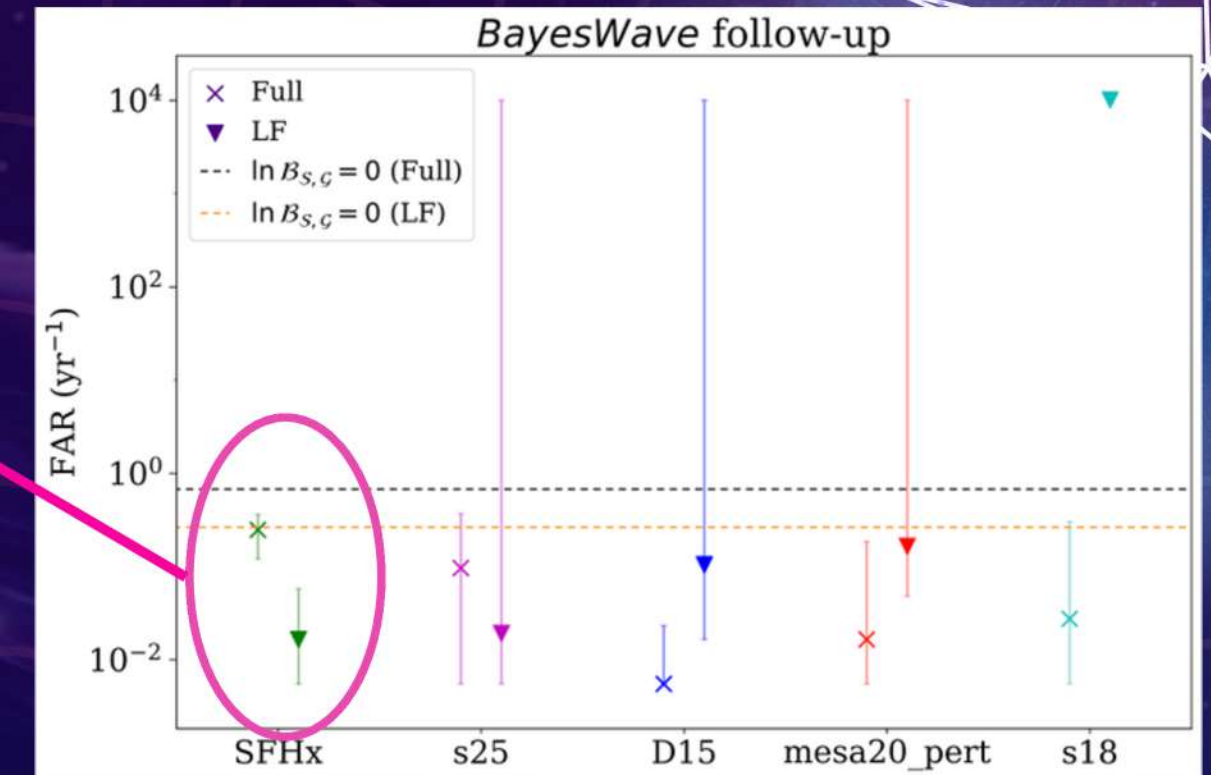


Decreasing LF power

Dedicated-frequency analysis **Summary**

CCSNe models with higher LF content have better detectability with LF follow-ups

But LF detectability is not guaranteed!
e.g. D15 and mesa20_pert



A successful LF detection is useful for constraining the CCSN explosion model,

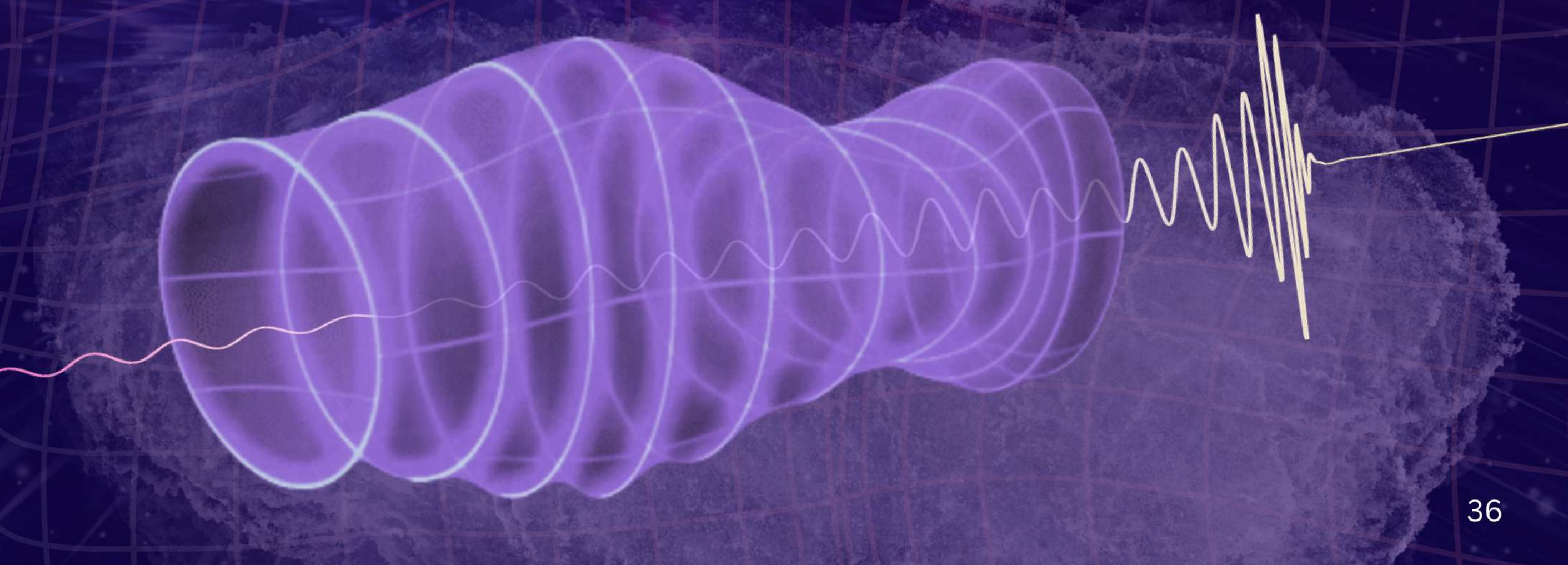
BUT...

Unsuccessful detection $\neq$ no LF emission

3

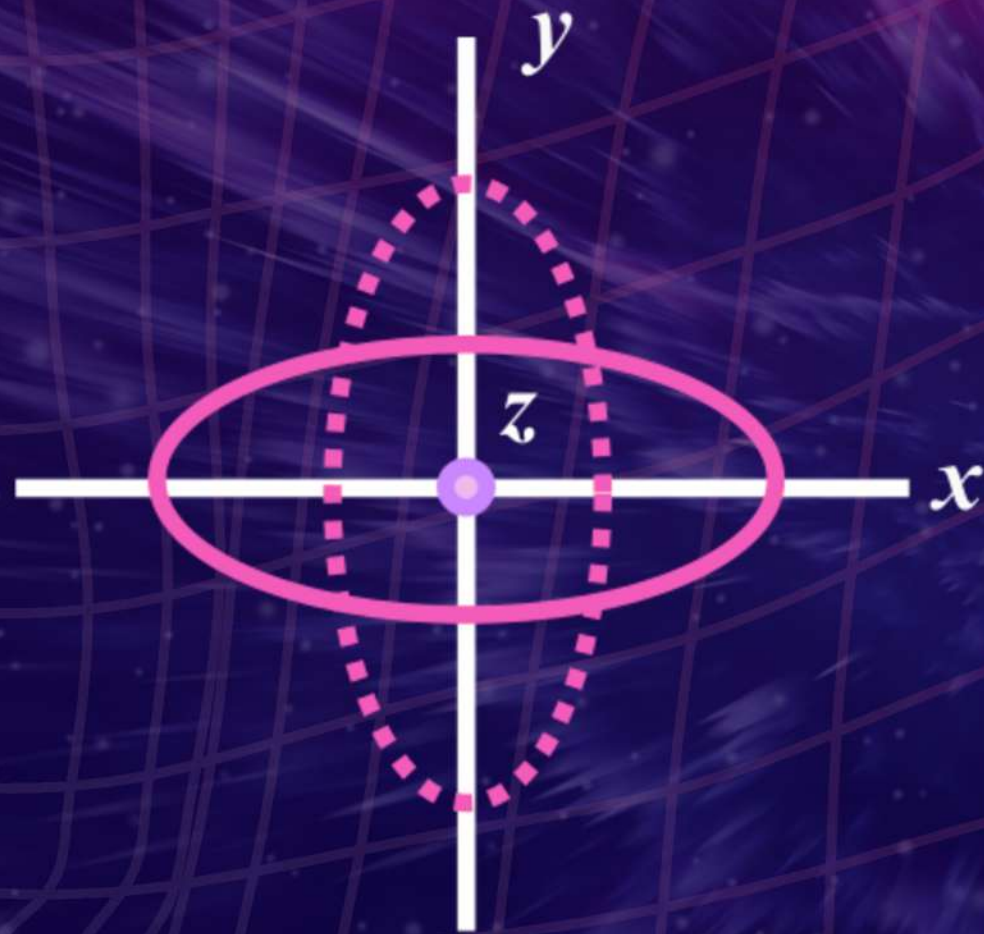
Characterizing
gravitational-wave burst

Tensor polarizations

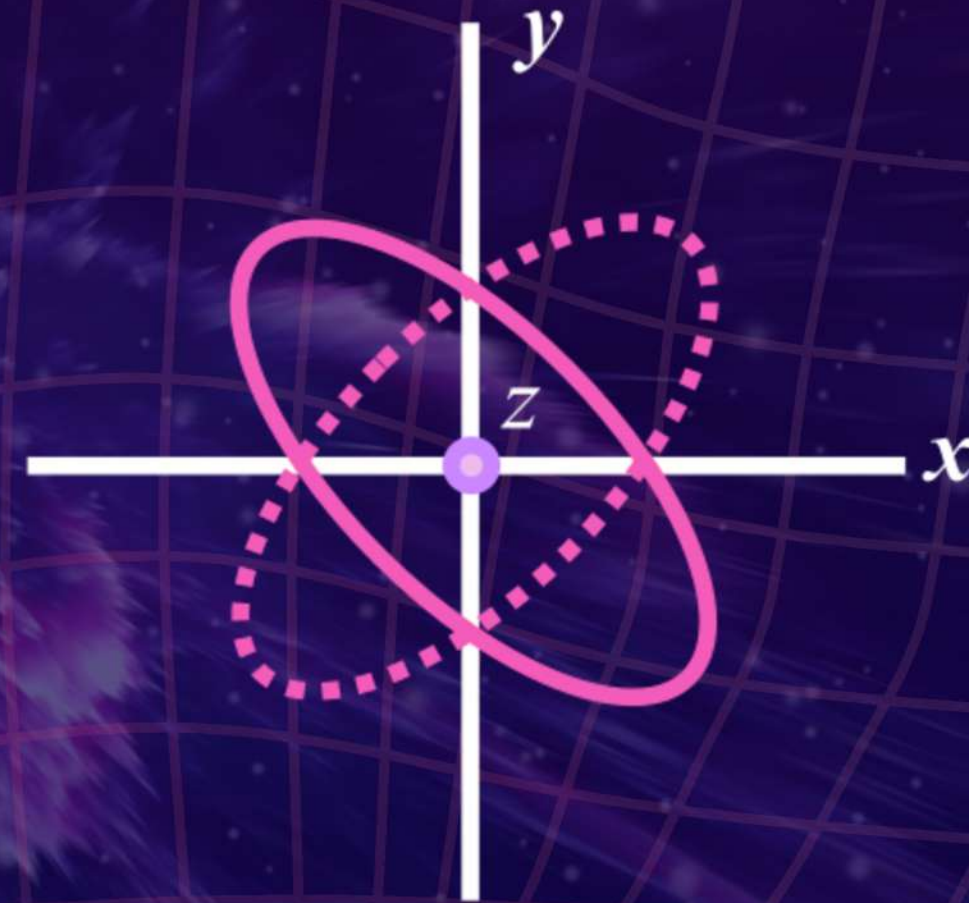


General relativity predictions

Tensor polarizations



Tensor plus (+)



Tensor cross ($\times$)

General Relativity predicts exactly
two polarization modes for GWs

General relativity predictions

Detector response

1 Frequency-domain response
of detector I

3 Time delay [phase]
correction factor

$$\tilde{h}_I = \left[F_{\times, I}(\Omega, \psi) \tilde{h}_{\times} + F_{+, I}(\Omega, \psi) \tilde{h}_{+} \right] e^{2\pi i f \Delta t_I(\Omega)}$$

2 **Linear combination** of plus [+] and cross [x] amplitudes;
 $F_{+}(\Omega, \phi)$ & $F_{\times}(\Omega, \phi)$ are the respective antenna pattern functions.

Signal polarization models

BayesWave

Elliptical polarization, E

Polarization amplitudes are restricted to

$$\tilde{h}^{\times} = i\epsilon\tilde{h}^{+}$$

ϵ is the ellipticity parameter, ranging from 0 (linear) to 1 (circular)

Relaxed polarization, R

Polarization amplitudes are reconstructed separately.

$$\tilde{h}^{\times} \text{ \& \& } \tilde{h}^{+}$$

but they share the same time-frequency content.

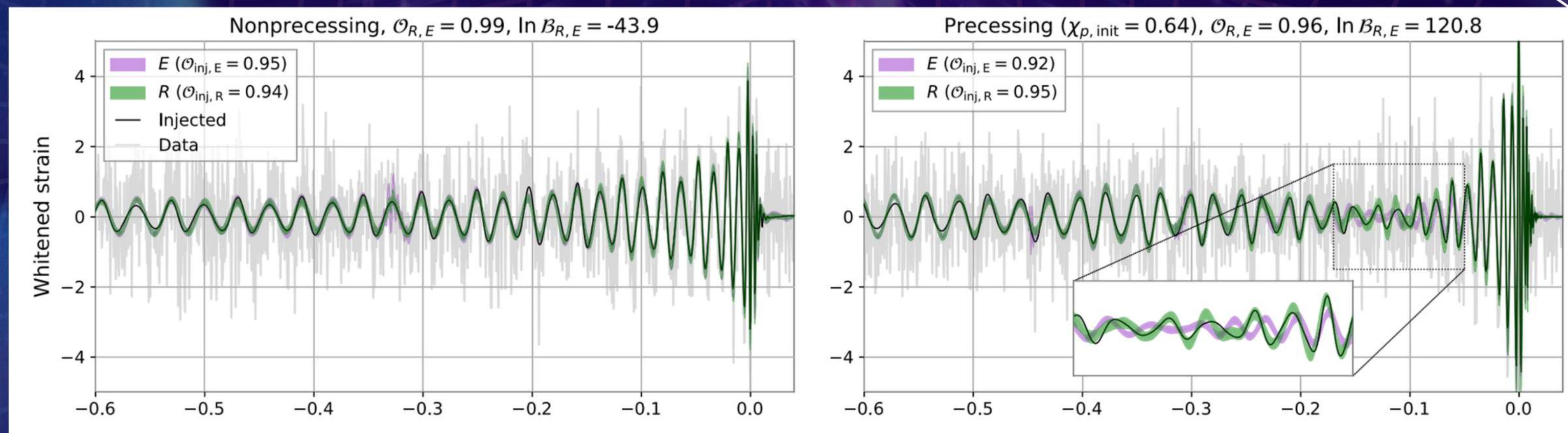
Why do we need R ?

E does not hold for compact binary coalescence signals with time-varying polarization

Examples:

Distinctive higher-order modes

Spin-precession (amplitude modulation, see below ↓)



Characterizing polarizations

Elliptical vs. Relaxed

How well do the E and R polarisation models represent elliptical and nonelliptical GW signals?

Is there a preferred model for different signal morphologies?

Is the preference affected by the size of detector network?

Injection set (Non-)precessing BBHs

HIGH MASS RATIO ($40M_{\odot} - 8M_{\odot}$) BBH WAVEFORMS INJECTED INTO
SIMULATED GAUSSIAN NOISE FOR HL, HLV AND HLKV

ELLIPTICAL
NON-PRECESSING ($\chi_p = 0$)

NON-ELLIPTICAL
PRECESSING ($\chi_{p,\text{init}} \neq 0$)

Elliptical vs. Relaxed

Figures of merit

Bayes factor

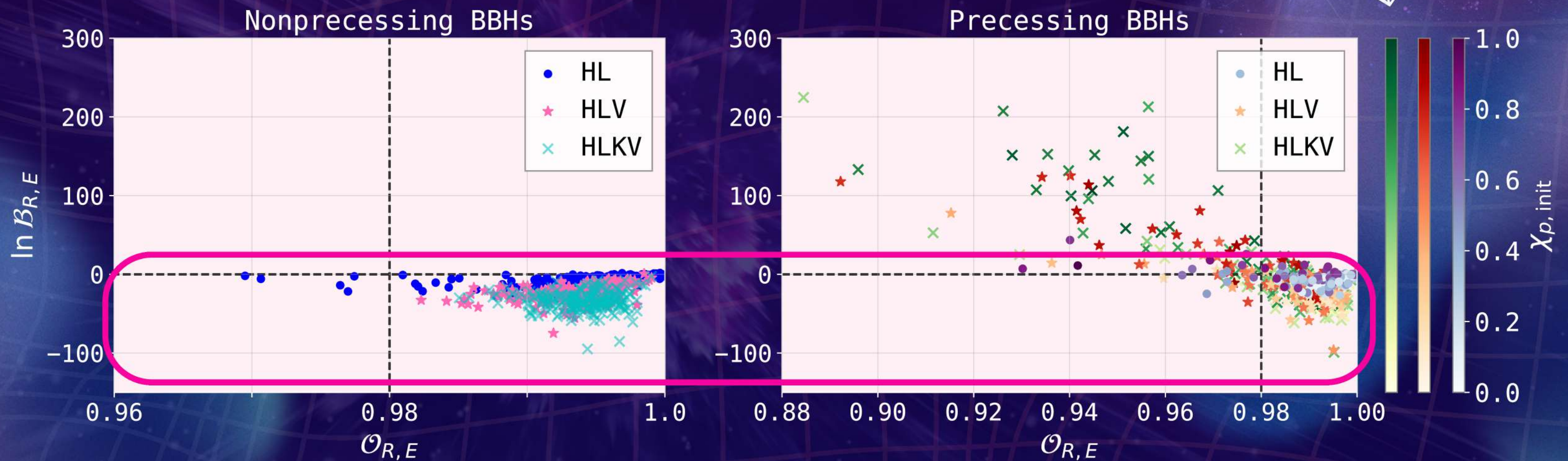
$$\ln \mathcal{B}_{R,E} = \ln(\text{Elliptical evidence}) - \ln(\text{Relaxed evidence})$$

Overlap between R & E reconstructions

$$0 \leq \mathcal{O}_{R,E} \leq 1 \quad \begin{cases} \rightarrow 0 = \text{no match} \\ \rightarrow 1 = \text{perfect match} \end{cases}$$

Elliptical vs. Relaxed

Performance comparison

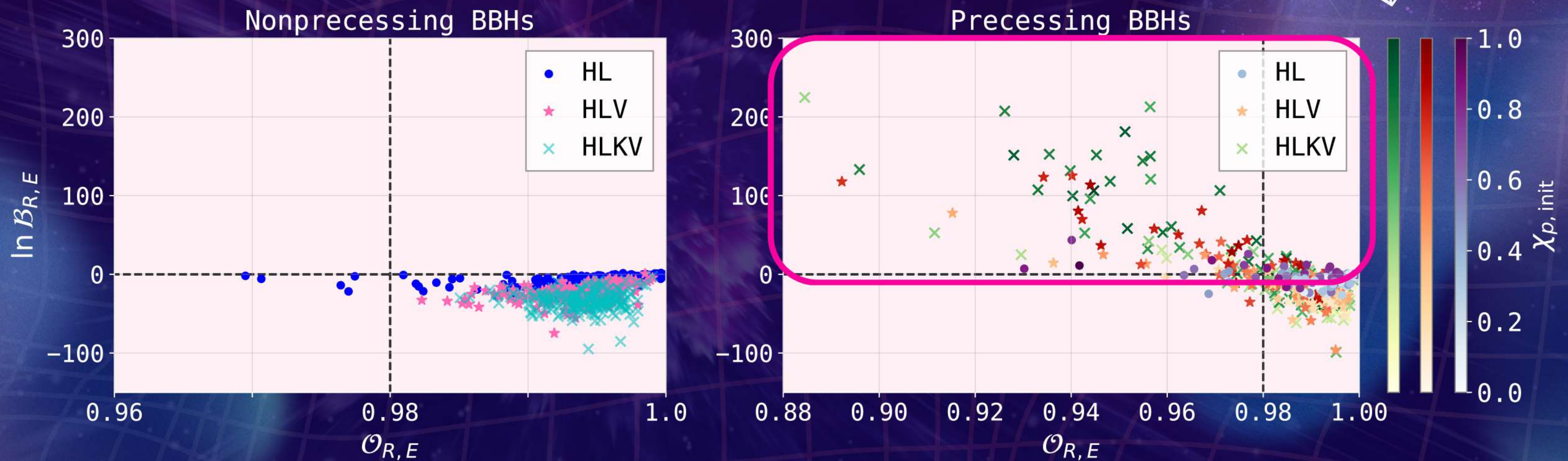


Non-precessing & minimally-precessing BBHs

$$\mathcal{O}_{R,E} > 0.98 \quad \ln \mathcal{B}_{R,E} < 0$$

Elliptical vs. Relaxed

Performance comparison



Strongly precessing BBHs

$$\mathcal{O}_{R,E} < 0.98$$

$$\ln \mathcal{B}_{R,E} > 0$$

Also increases
with larger
detector networks

Elliptical vs. Relaxed

Key takeaway

$\ln \mathcal{B}_{R,E}$ and $\mathcal{O}_{R,E}$
together can provide indications for
nonelliptical polarization,
but are insufficient to distinguish between
purely elliptical and slightly nonelliptical
signals.

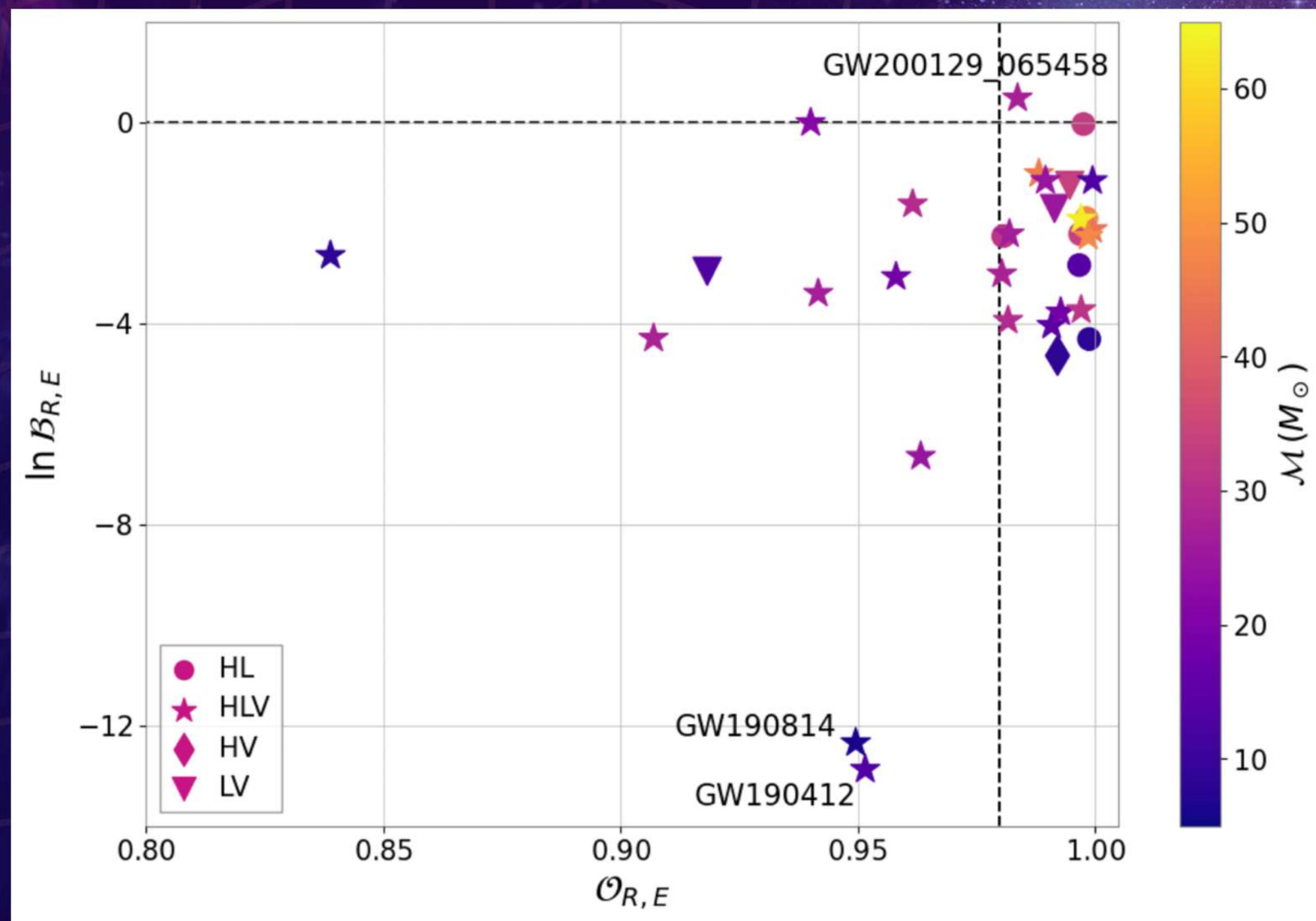
Elliptical vs. Relaxed with real O3 GW events

$$\mathcal{O}_{R,E} > 0.90$$

E and R
reconstructions are
comprable

$$\ln \mathcal{B}_{R,E} < 0$$

O3 events are
consistent the elliptical
polarization model



R model reconstruction of Stokes parameters

Stokes parameters [I, Q, U, and V] are linear combinations of $\tilde{h}^\times$ & $\tilde{h}^+$ which describe the polarization content of a GW signal.

Fractional polariztions

Total

$$F_T(f) = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}$$

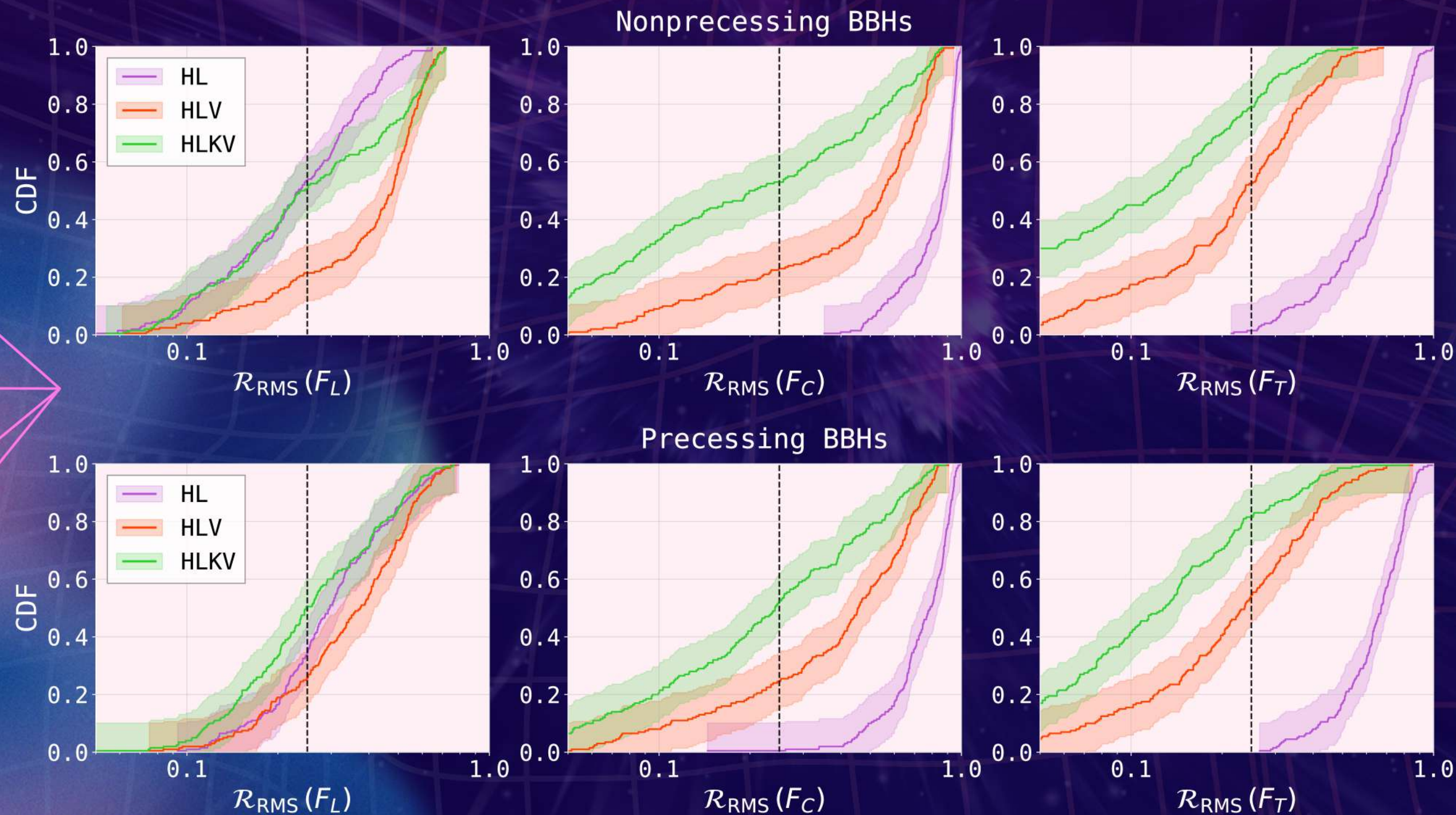
Circular

$$F_C(f) = \frac{|V|}{I}$$

Linear

$$F_L(f) = \frac{\sqrt{Q^2 + U^2}}{I}$$

Accuracy of Polarization measurements



Cumulative distribution functions of root mean square residuals $\mathcal{R}_{\text{RMS}}$ between the injected and recovered fractional polarizations

Key takeaway

1. Accuracy improves with larger networks
2. Signal morphology does not affect the accuracy

Characterizing polarizations

Summary

E is generally preferred for its simplicity compared to R

but R is preferred when the signal clearly exhibits time-varying polarization content

R can be used to measure polarization content of GW signals in terms of Stokes parameters

performance improves with expanded detector networks

OPEN ASTROPHYSICS CODEBASES

DEVELOPED AND MAINTAINED BY

THE UNIVERSITY OF MELBOURNE GROUP



NEUTRON STAR THEORY

Quantum fluids

Vortex motion:

Gross-Pitaevskii solver
N-body solver

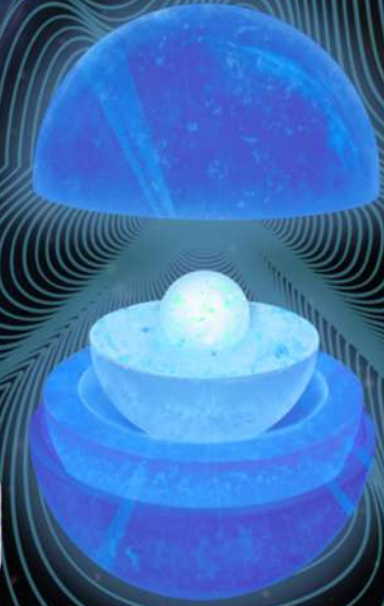
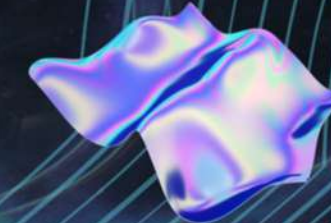
Thermal evolution model for hyperon stars

Neutron star oscillations:
Numerical relativity solver

Magnetic fields

Magnetic mountains:

Grad Shafranov solver
General relativistic formulation



PULSAR TIMING

Audioband GWs (LVK)

Continuous-wave [CW] detection:

Hidden Markov model + Viterbi

Adaptive noise cancellation:

Recursive least squares



Nanohertz GWs (PTA)

CW parameter estimation for SMBH binaries:

Kalman filter

GW background detection:

Kalman filter



Radio

Pulsar glitch detection

Hidden Markov model

Pulsar detection in compact binaries

Hidden Markov model

Two-component pulsar parameter estimation

Kalman filter

Modelling of young pulsar secular spin-down and timing-noise

Brownian framework

X-ray

Low-mass X-ray binary parameter estimation

Kalman filter



OZGRAV

ARC CENTRE OF EXCELLENCE FOR GRAVITATIONAL WAVE DISCOVERY

Education & Outreach

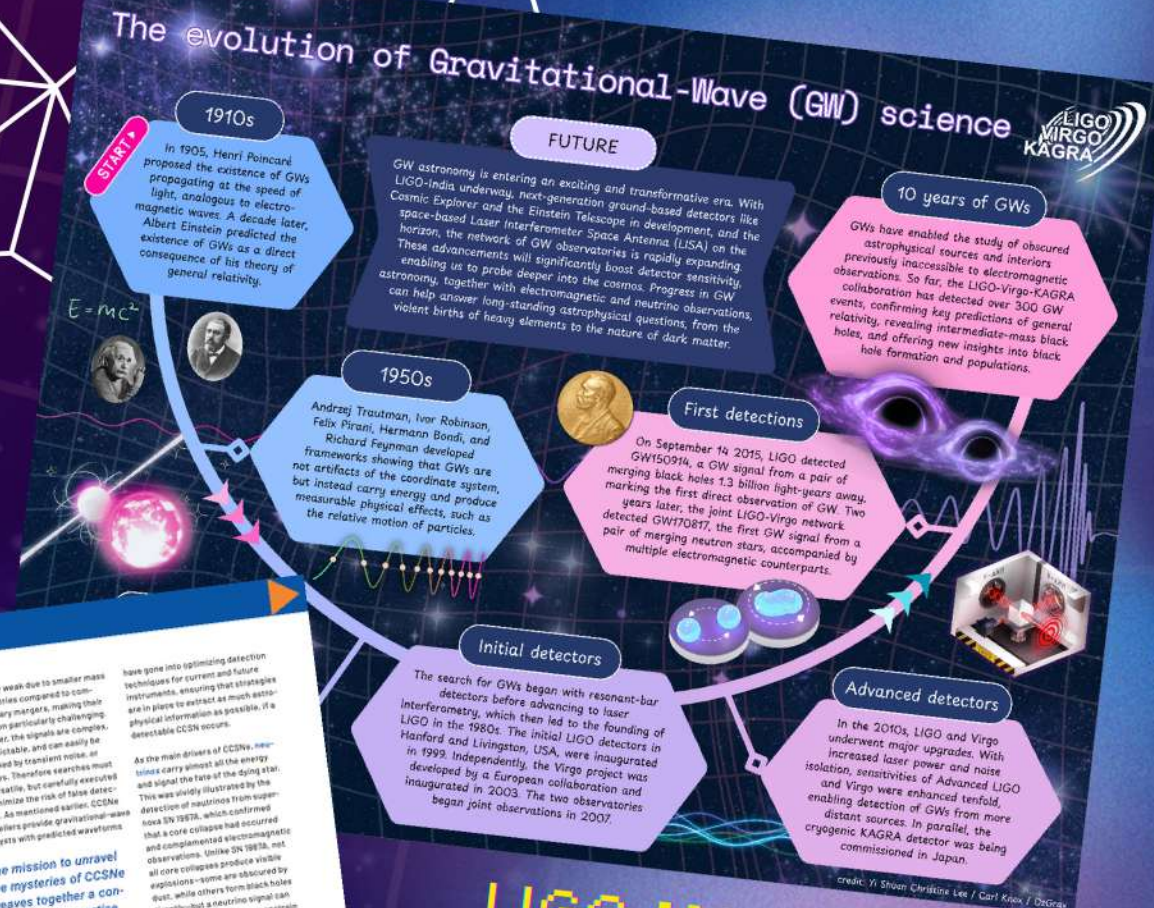


OZGRAV OUTREACH EFFORTS



VR SYSTEM

OUTREACH AMBASSADOR PROGRAM



LIGO MAGAZINE

Toward Unraveling the Mysteries of CCSNe

Hunting core-collapse supernovae: A unified effort

Massive stars, with more than eight times the mass of the Sun, end their lives in a Core-Collapse Supernova (CCSN). In this process, the star's core collapses inward, and the outer layers are ejected, creating a shock wave that propagates outward, generating a bright explosion. Despite their brevity, CCSNe leave lasting imprints, forging the elements that make up much of everything and everyone we know, and thus hold the answers to our cosmic origins.

CCSNe have been observed and studied in various ways for millennia. Chinese astronomers charted the paths of "guest stars," and today they are studied using modern telescopes and instruments. Yet CCSNe have remained elusive, with many questions still unanswered. This article highlights the LIGO-Virgo-KAGRA collaboration, alongside the wider astronomy and astrophysics community, combines diverse expertise to advance this long-standing quest.

Modelling CCSNe
CCSNe involve a complex interplay of physical processes. Studying how these processes influence one another relies on detailed computer simulations. However, a type of virtual modelling, known as multi-messenger observations, provides a window into the hidden depths of CCSNe. Electromagnetic observations complement these multi-messenger signals by providing both the time window and the location for targeted follow-up searches, particularly for gravitational waves.

Multi-messenger observations
The first messenger through which CCSNe were observed is, of course, electromagnetic radiation. Hundreds of CCSNe have been observed, providing insights into the explosion energetics, the dynamics of the ejecta, and the formation of the remnant. However, electromagnetic signals cannot generate magnetic fields or reveal the interior of the star, and thus cannot reveal the full picture of the explosion. Fortunately, the use of multi-messenger astronomy is upon us: the largely unexplored propagation of neutrinos and gravitational waves provides a new observational window into the hidden depths of CCSNe. Electromagnetic observations complement these multi-messenger signals by providing both the time window and the location for targeted follow-up searches, particularly for gravitational waves.

Gravitational waves from CCSNe
Gravitational waves from CCSNe are expected to be relatively weak due to smaller mass asymmetries compared to compact binary mergers, making their detection particularly challenging. However, the signals are complex, non-repeating, and can easily be mimicked by transient noise or glitches. Therefore, searches must be sensitive to a wide range of possible signals, but also carefully executed to minimize the risk of false detections. As mentioned earlier, CCSNe have been observed to produce gravitational waves, and this provides a unique window into the explosion's interior dynamics. Accurate models are also crucial for predicting multi-messenger signals: predicting electromagnetic, neutrino, and gravitational wave signals from the explosion energetics, the dynamics of the ejecta, and the formation of the remnant. These signals bear the imprint of the supernova's central engine.

The mission to unravel the mysteries of CCSNe
As detecting multi-messenger signals remains a formidable challenge—limited by instrument sensitivity and the rarity of CCSNe in our galaxy—success depends on preparation, ingenuity, and a touch of cosmic luck.

Reflection
The mission to unravel the mysteries of CCSNe is a complex one, requiring a combination of expertise. As detecting multi-messenger signals remains a formidable challenge—limited by instrument sensitivity and the rarity of CCSNe in our galaxy—success depends on preparation, ingenuity, and a touch of cosmic luck.

Reflection
The mission to unravel the mysteries of CCSNe is a complex one, requiring a combination of expertise. As detecting multi-messenger signals remains a formidable challenge—limited by instrument sensitivity and the rarity of CCSNe in our galaxy—success depends on preparation, ingenuity, and a touch of cosmic luck.

Laboratory and Computational Physics 3: Bayesian Inference and Gravitational-wave data analysis

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THIRD YEAR
GW LAB
ACTIVITY
@ UNIMELB

Thank You

Come chat with me!

- GW burst detection & characterisation
- Core-collapse supernovae
- Detector characterization - continuous waves
- Pulsar glitch detection with Hidden Markov models
- Open Astrophysics Codebase (Uni of Melbourne)
- OUTREACH AND TEACHING (AND CATS!)

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